Algorithms for the Analysis of Biological Sequences 2025/2026

Molecular Evolution and Phylogenetics

Miriam Santos
[dCC] @ Faculty of Sciences University of Porto

Outline

- **Recovering Evolutionary History**
  - Phylogenetic Trees and Evolutionary History
  - Gene and Species Trees
  - Phylogenetic Tree Structure
  - Types of Phylogenetic Trees
  - Tree topology and Newick format
  - Practical Examples

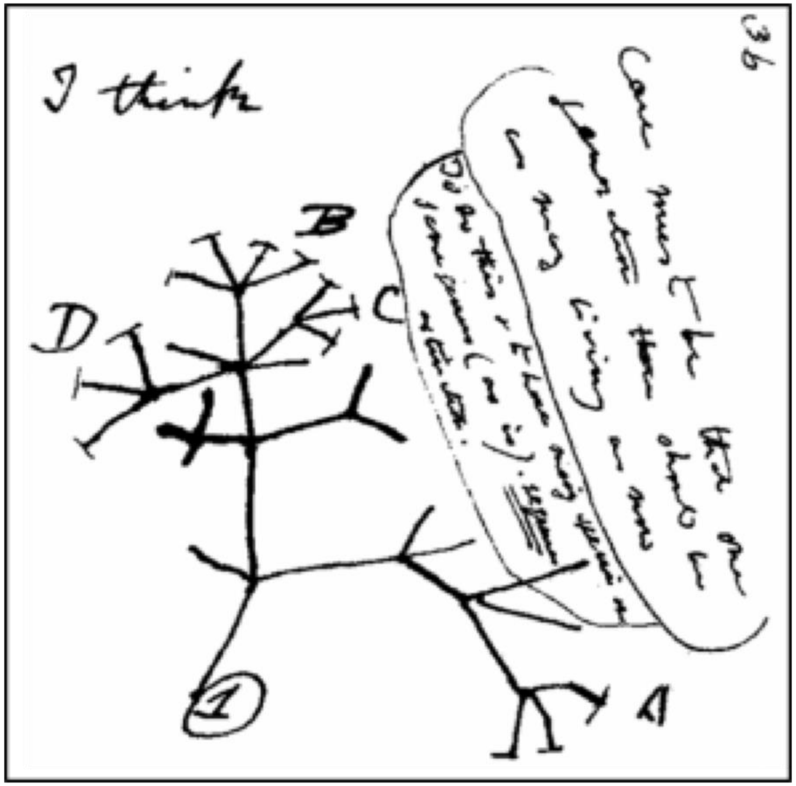
- **Building Phylogenetic Trees**
  - Algorithms for Phylogenetic Analysis
  - UPGMA and Neighbourhood Joining Methods
  - Biopython for Phylogenetic Analysis
  - Practical Examples

Phylogenetics

- Phylogenetics is the branch of Biology which studies the **evolutionary history and relationships** among individuals or species.

- While philogenies have been mainly studied based on heritable, morphological traits, **the availability of DNA sequences has reshaped the field**, given that mutations occur directly over the DNA.

- In the context of Bioinformatics, phylogenetic analysis is concerned with **determining how a given set of sequences has evolved from a common ancestor** through a process of natural evolution, depicted as an evolutionary tree.



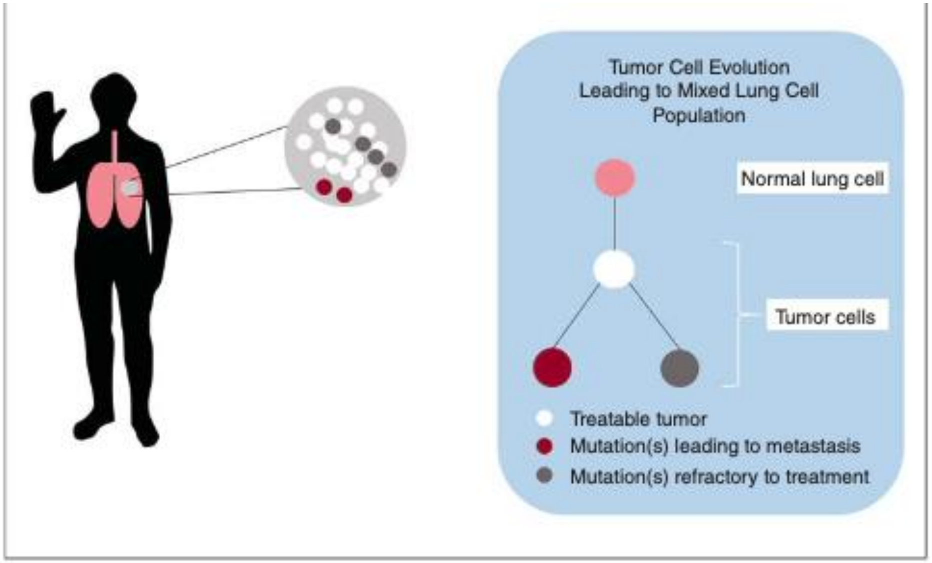
Sketch by Charles Darwin using the metaphor of a tree to represent evolutionary relationships.

Phylogenetics Applications

- There are multiple applications of phylogenetics such as **classifying new species**, **forensics** (fatherhood determination, identifying individuals commiting crimes, fighting bioterrorism, e.g., anthrax), **epidemiology (**e.g., tuberculosis, salmonella, e. coli, zika**)** and **conservation policies** (endangered species),

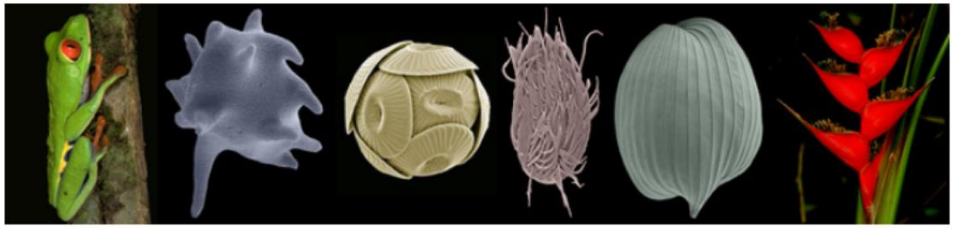
Box 9.1 Predicting Cancer Progression and Drug Response Using Phylogenetic Approaches

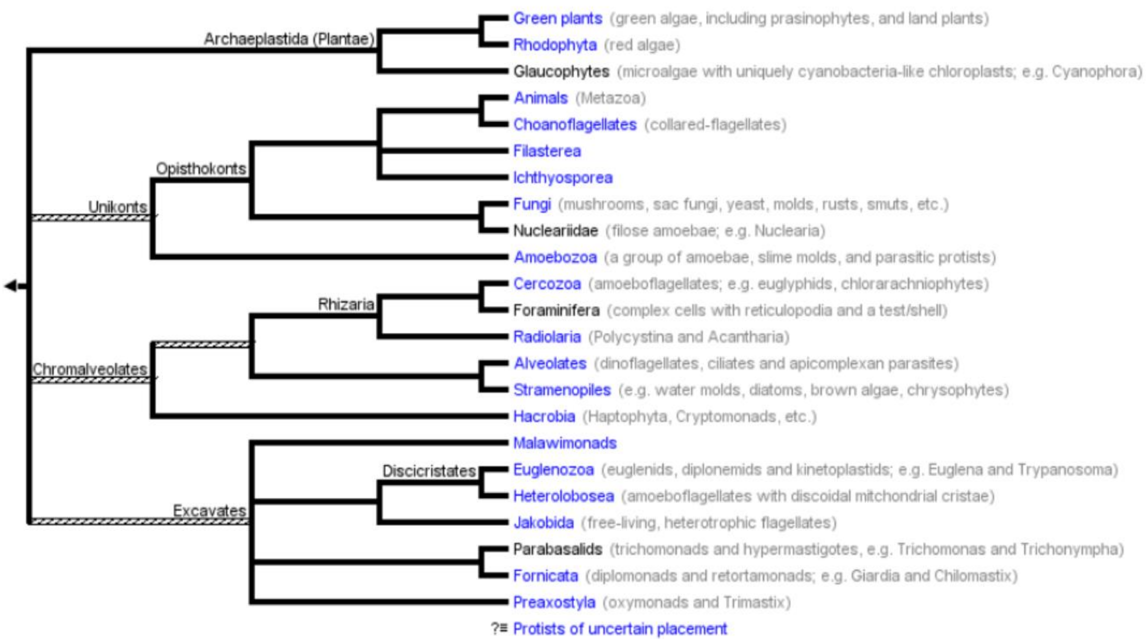
Tumor phylogenetics provides insight into evolutionary mechanisms causing disease. Cancer is a genetic disease in which the accumulation, diversification, and selection for mutations are now known to promote tumor cell proliferation and impact survival according to complex evolutionary mechanisms. The figure shows how tumors can contain a mixed population of cells that have accumulated different types of mutations. Some mutations enable cancer cells to metastasize, while others render cancer cells less susceptible to treatment. Phylogenetic analysis has been applied to the understanding of predicting and controlling cancer progression, metastasis, and therapeutic responses. Tumor phylogenetics aims to reconstruct tumor evolution from genomic variations by exploring the space of possible trees in order to explain a dataset. In particular, evolutionary theory and analyses have been developed for determining the heterogeneity of tumor cells, specifically the types of mutations associated with different clinical outcomes; these include copy number variants, microsatellites (tracts of repetitive DNA in which certain DNA motifs are repeated), and "mutation signatures" such as nucleotide biases linked to environmental triggers. Rates of mutation, as well as the extent and intensity of selective pressures, greatly affect treatment options and prognosis. Different treatment regimens lead to selection that can, in turn, alter the dominant clones within tumors. Single-agent treatment can lead to relapse by selecting for non-responsive clones and higher mutation rates (intratumor heterogeneity), and has been linked with the ability to resist different types of therapy. These types of tumor diversity studies depend highly on appropriate parameter estimation and modeling of different mutational processes that have been validated with observed data. Most studies of tumor phylogenetics to date have adapted standard algorithms that were developed for generating phylogenies of different species (Schwartz and Schaffer 2017).



The Tree of Life: tolweb.org

- There are three major **reasons to study phylogenetics:**
  - Determining the **closest relative** of the organism we are interested in
  - Discovering the **function** of a gene
  - Retracing the **origin** of a gene





Homologs, Orthologs and Paralogs

- **Homologs are genes that share a common evolutionary history. They descend from the same ancestor.**
  - They can be orthologs or paralogs.

- **Orthologs are homologous genes found in different species that derive from a common ancestor and separated because of speciation.**
  - They usually keep similar function (have similar roles)
  - Human alpha-globin <> Mouse alpha-globin

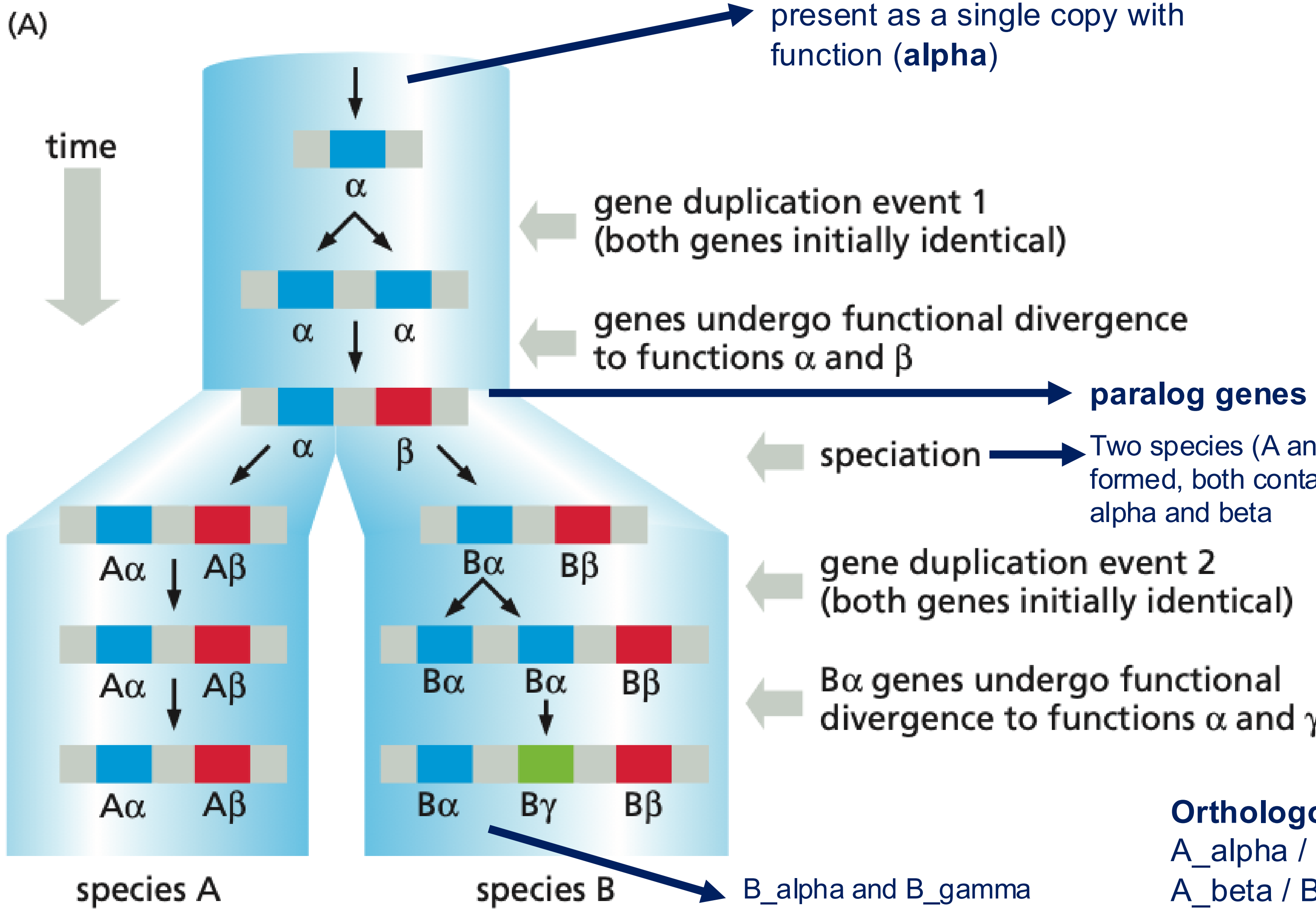
- **Paralogs are homologous genes generated from a gene duplication event, usually withing the same genome or lineage.**
  - They often evolve into new of specialized functions
  - Human alpha-globin <> Human beta-globin

- **Ortho = “other species”**
- **Para = “parallel copies” in a genome**

Homologs, Orthologs and Paralogs

All 5 genes in both species are **homologous**.

In the ancestral species, a gene is present as a single copy with function (**alpha**)



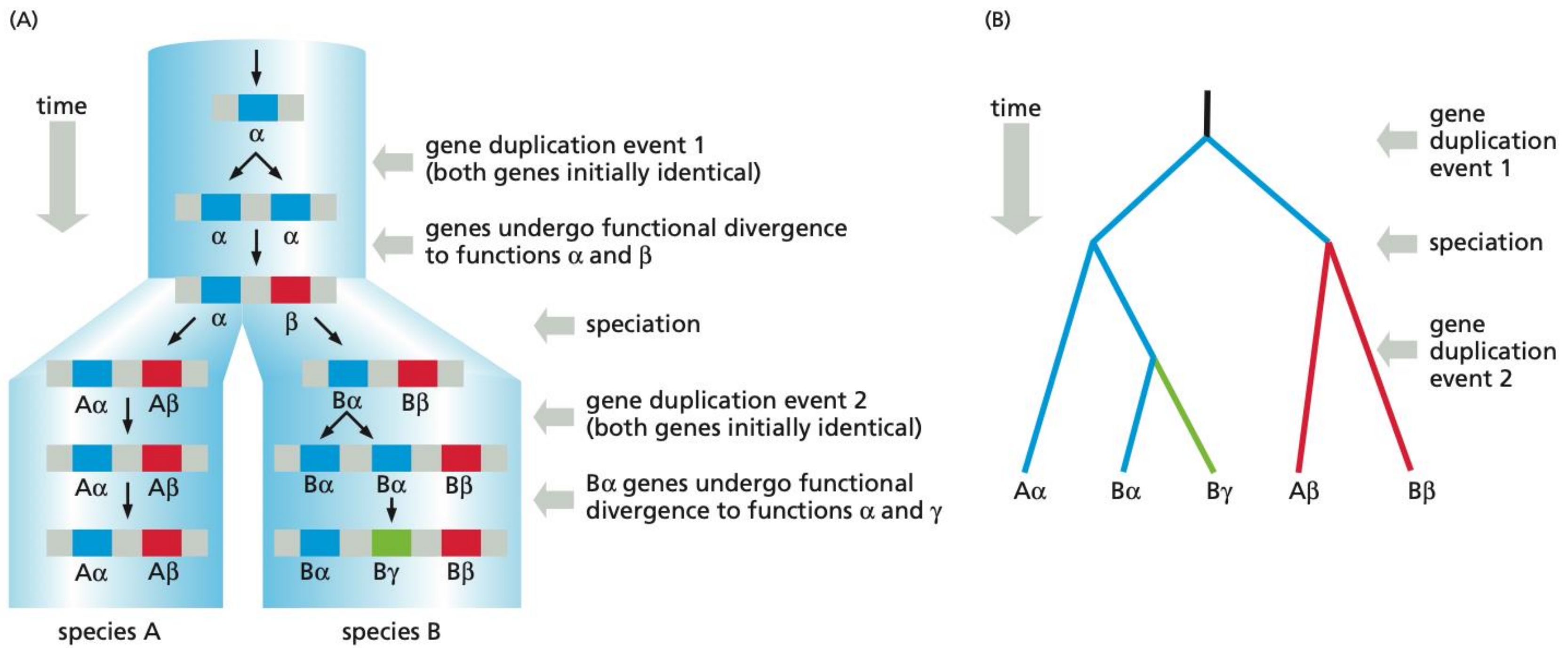
paralog genes

Two species (A and B) are formed, both containing genes alpha and beta

B_alpha and B_gamma are paralogs

Orthologous pairs:
A_alpha / B_alpha
A_beta / B_beta
A_alpha / B_gamma

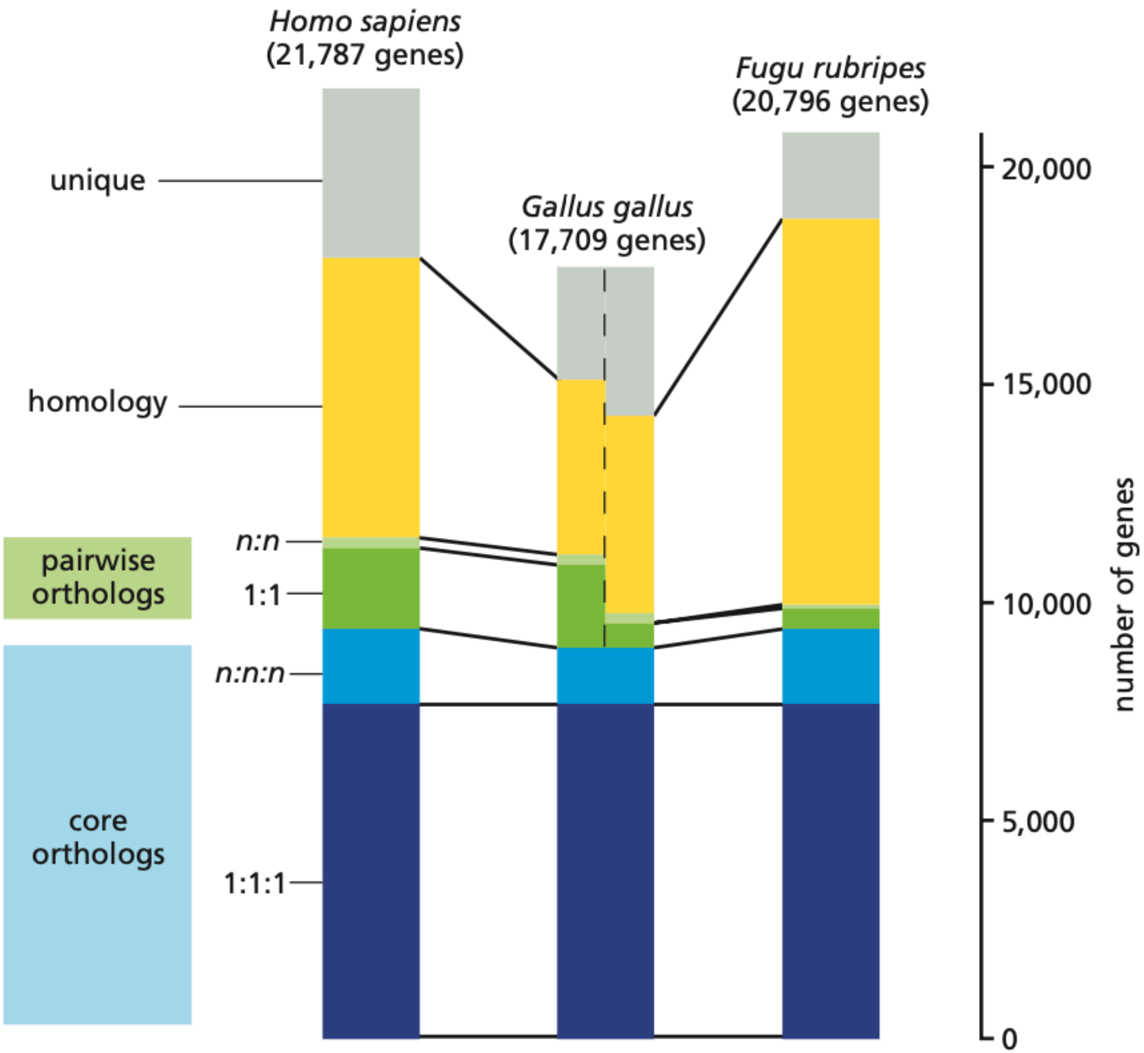
Homologs, Orthologs and Paralogs



The evolutionary history of a gene that has undergone two separate duplication events. (A) A species tree is depicted by the pale blue cylinders, with the branch points (nodes) in the cylinders representing speciation events. In the ancestral species a gene is present as a single copy and has function α (blue). At some time a gene duplication event occurs within the genome, producing two identical gene copies, one of which subsequently evolves a different function, identified as β (red). These are paralogous genes. Later a speciation event occurs, resulting in two species (A and B) both containing genes α and β. Gene Bα (in species B) subsequently undergoes another duplication event, which after further divergent evolution results in genes Bα and Bγ, the latter with a new function γ (green). The Bα gene is still functionally very similar to the original gene. At the end of this period of evolution, all five genes in both species are homologous, with three orthologous pairs: Aβ/Bβ, Aα/Bα, and Aα/Bγ. The Bα and Bγ genes are paralogous, as are any other combinations except the orthologous pairs. Note that Aα and Bγ are orthologs despite their different functions, and so if the intention is to study the evolution of a particular functional product, such as the α function, we need to be able to distinguish the Aα/Bα pair from the Aα/Bγ pair. This can be done using sequence similarity, which would be expected to be greater for the Aα/Bα pair as they will be evolving under almost identical evolutionary pressures. Errors in functional orthology assignment can easily occur, depending on sequence and functional similarity and whether all related genes have been discovered. (B) The phylogenetic tree that would be drawn for these genes, here drawn as a cladogram.

Homologs, Orthologs and Paralogs

The numbers of orthologous, homologous, and unique genes in the human (*Homo sapiens*), chicken (*Gallus gallus*), and puffer fish (*Fugu rubripes*) genomes. Orthologs present as one copy in each of the three species are represented in dark blue on the histogram and are labeled 1:1:1; genes represented in lighter blue and labeled n:n:n are orthologous genes present in all three species, which have been duplicated in at least one of the species. These two groups are considered as core orthologs for the three species. Orthologs (and duplicated orthologs) found in only two species are labeled pairwise orthologs, represented as 1:1 (or n:n) and colored green. Homologous genes for which no clear orthologous relationships can be determined in the pairwise comparison are in yellow, and unique genes are in gray. (Reprinted



Gene or Species Trees

- **If you select a group of homologous genes, you build a gene tree.**

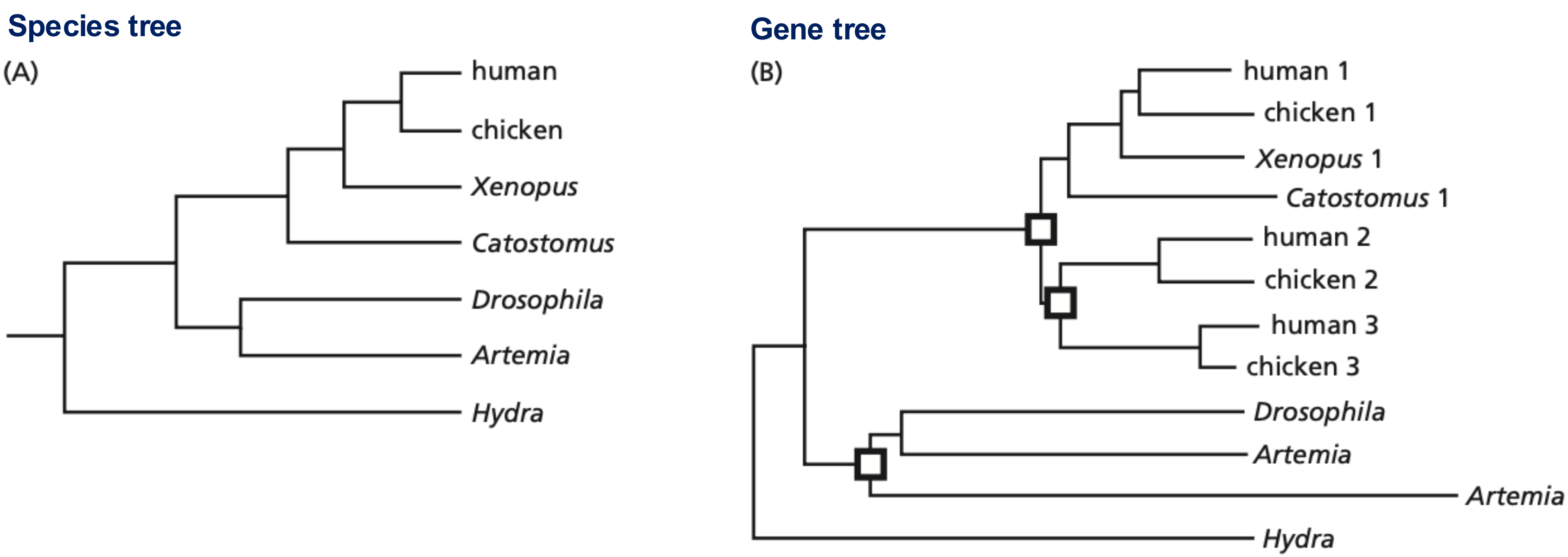
  - If you select all ***paralogous*** members of a large human gene family, your gene tree tells the story of this **gene family** only. You can reconstruct the chain of **duplications that led from one single ancestral gene to the current situation**.

  - If you select a group of genes that are all ***orthologous*** from different species, the gene tree looks like a **species tree**. You can reconstruct the **speciations that occurred** while the species were diverging.

- When examining a phylogenetic tree, you should **note exactly what information was used to produce it**, because the evolutionary history of a set of related genes is often not the same as that of the species from which the genes were selected.

Gene or Species Trees

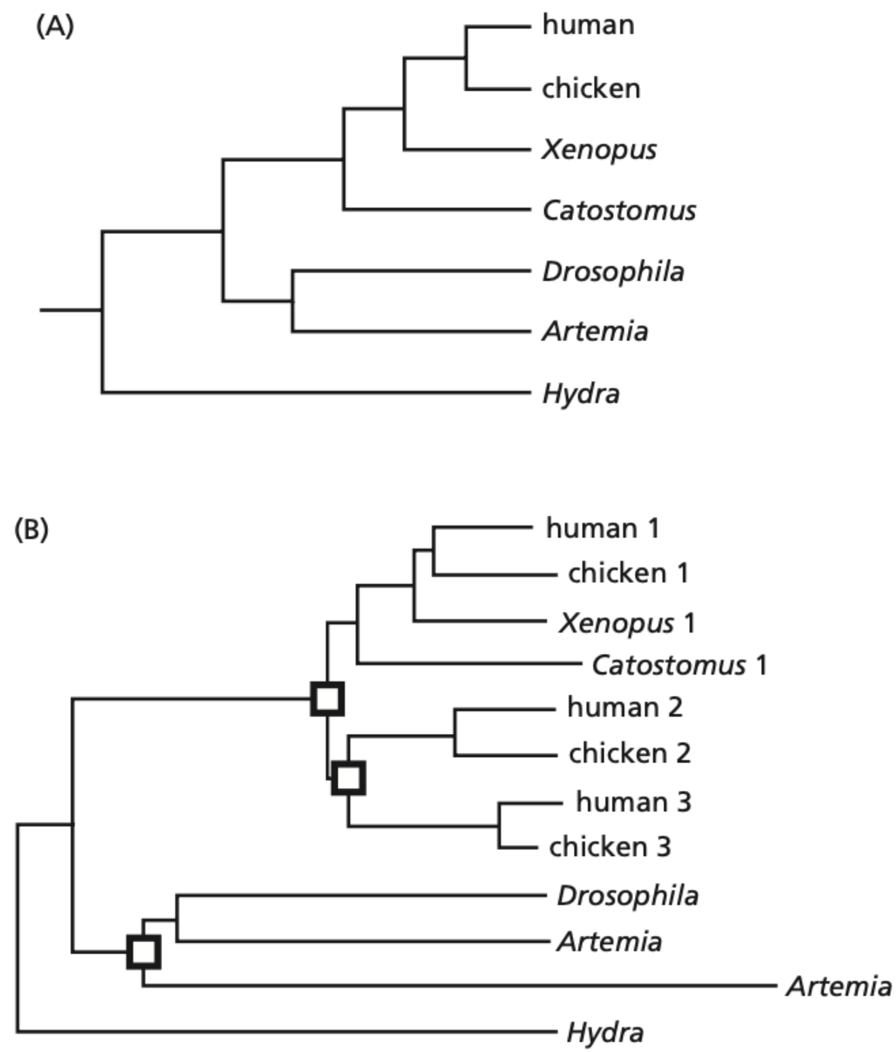
- It is important to be aware of the distinction between sequence-based phylogenetic trees intended primarily to **show evolutionary relationships between species** and those intended to **show relationships between members of a family of homologous genes or proteins from different species.**



A **gene family tree** charts the way in which an ancestral gene has become **duplicated and reduplicated** within genomes, giving rise to **distinct gene subfamilies**, and how the **copies have diverged** within the same species as well as between species. In gene family trees, some branch points represent gene **duplication** events within the same species while others represent **speciation** events.

Gene or Species Trees

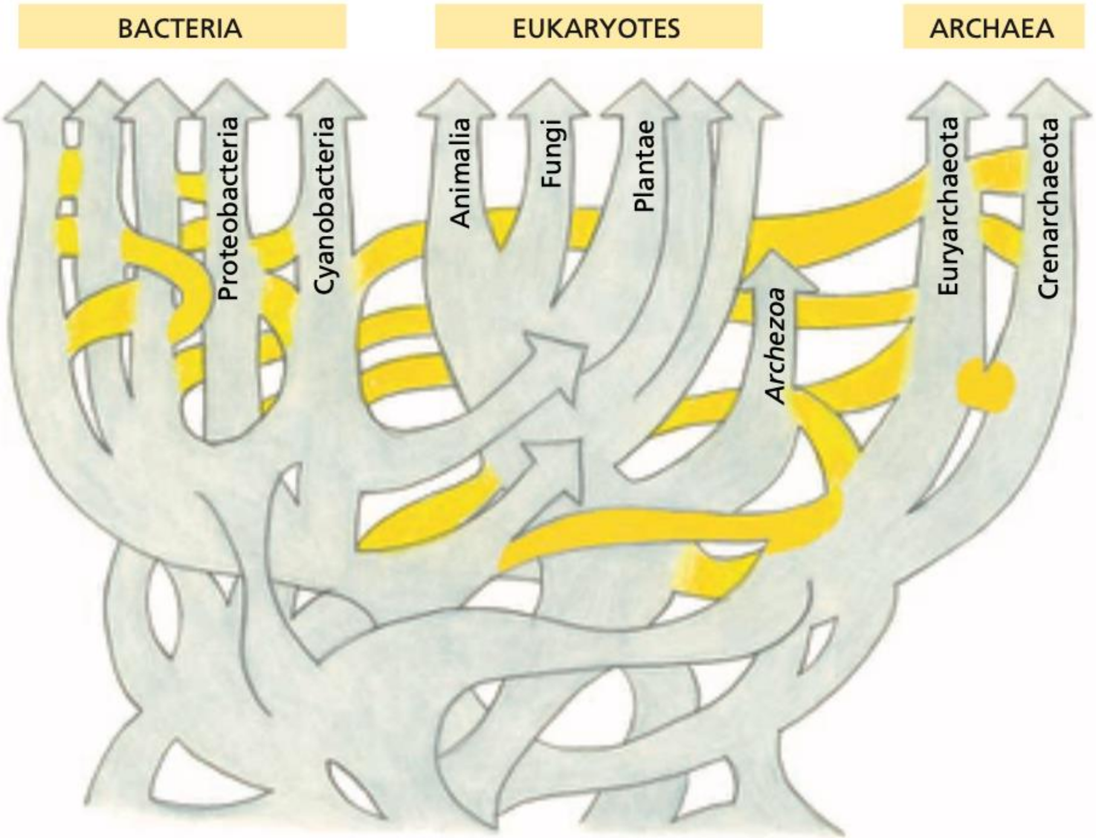
Examples of a species tree and a gene tree. (A) A species tree showing the evolutionary relationships between seven eukaryotes, with one more distantly related to the others (*Hydra*) used as an outgroup to root the tree. *Xenopus* is a frog, *Catostomus* a fish, *Drosophila* a fruit fly, and *Artemia* the brine shrimp. (B) The gene tree for the Na^+–K^+ ion pump membrane protein family members found in the species shown in (A). In some species, e.g., *Hydra*, and *Xenopus*, only one member of the family is known, whereas other species, such as humans and chickens, have three members. The small squares at nodes indicate gene duplications,



Horizontal Gene Transfer

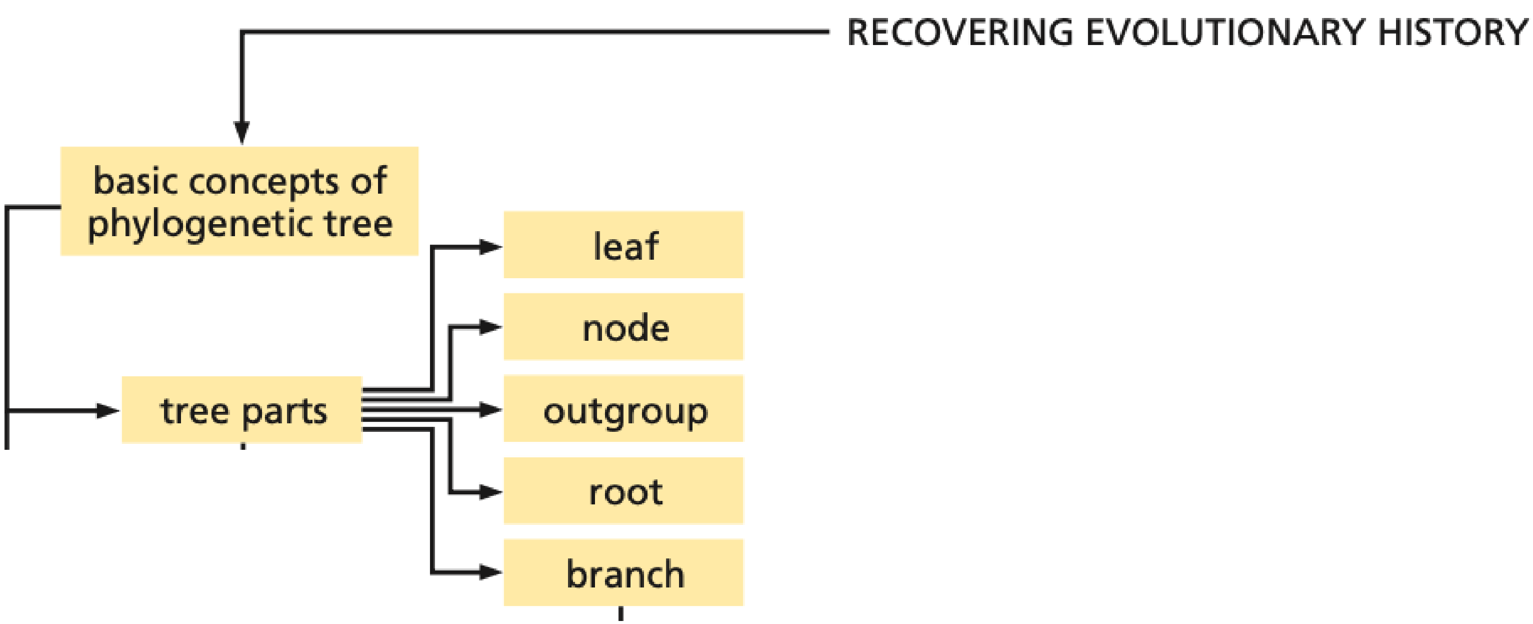
- Another evolutionary event that can confuse phylogenetic analysis is the process of **horizontal gene transfer (HGT),** also known as **lateral gene transfer (LGT): it involves a gene from one species being transferred into another species** *(e.g., by piggy-back-rinding a virus infection, for instance).*

A sketch of the tree of life including a considerable amount of horizontal gene transfer (HGT). HGT is distinguished from normal vertical gene transmission by yellow shading. The tree should not be interpreted too precisely, but does show that Eukaryotes are largely unaffected by HGT except in their earliest evolutionary period, whereas the Archaea and Bacteria continue to experience these events. (Adapted from W.F. Doolittle, Phylogenetic classification and the universal tree, *Science* 284:2124–2128, 1999.)



- A phylogenetic tree is a **diagram that proposes an hypothesis for the reconstructed evolutionary relationships** between a set of objects (which provide the data from which the tree is constructed).

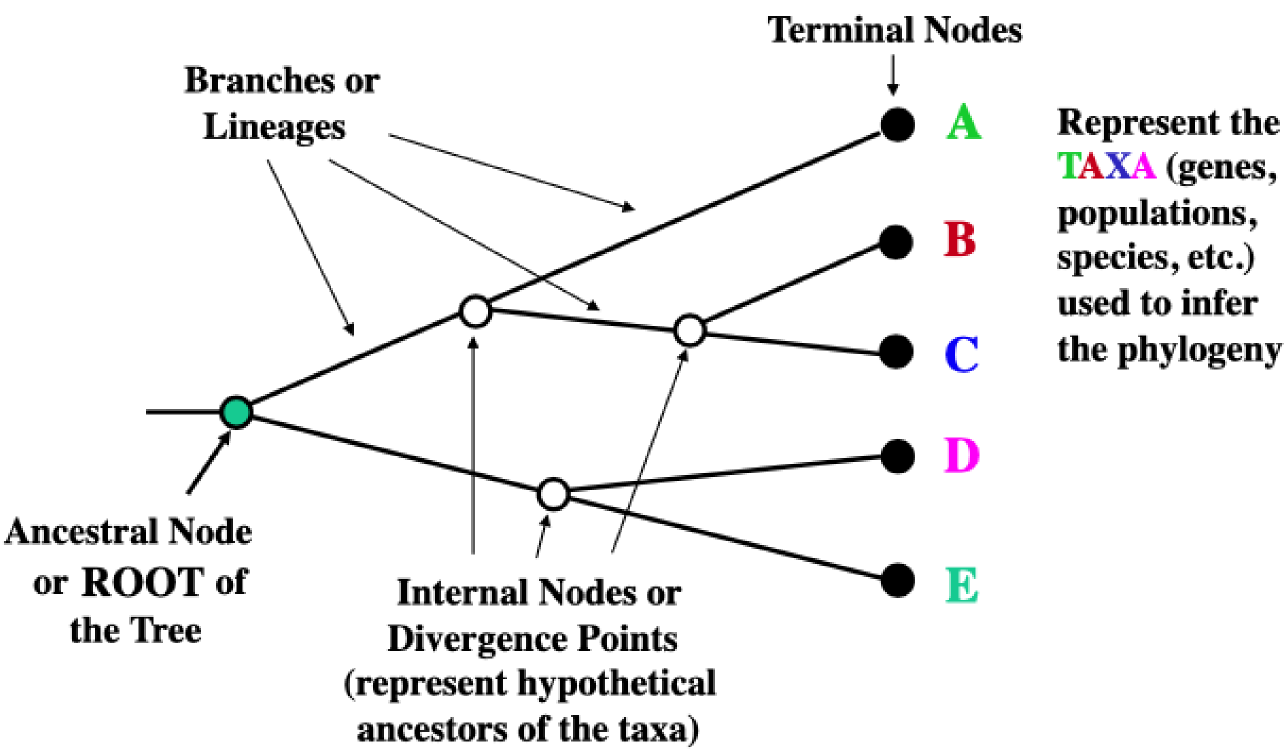
- When working with phylogenetic trees, there is "special vocabulary" involved:



Phylogenetic Trees

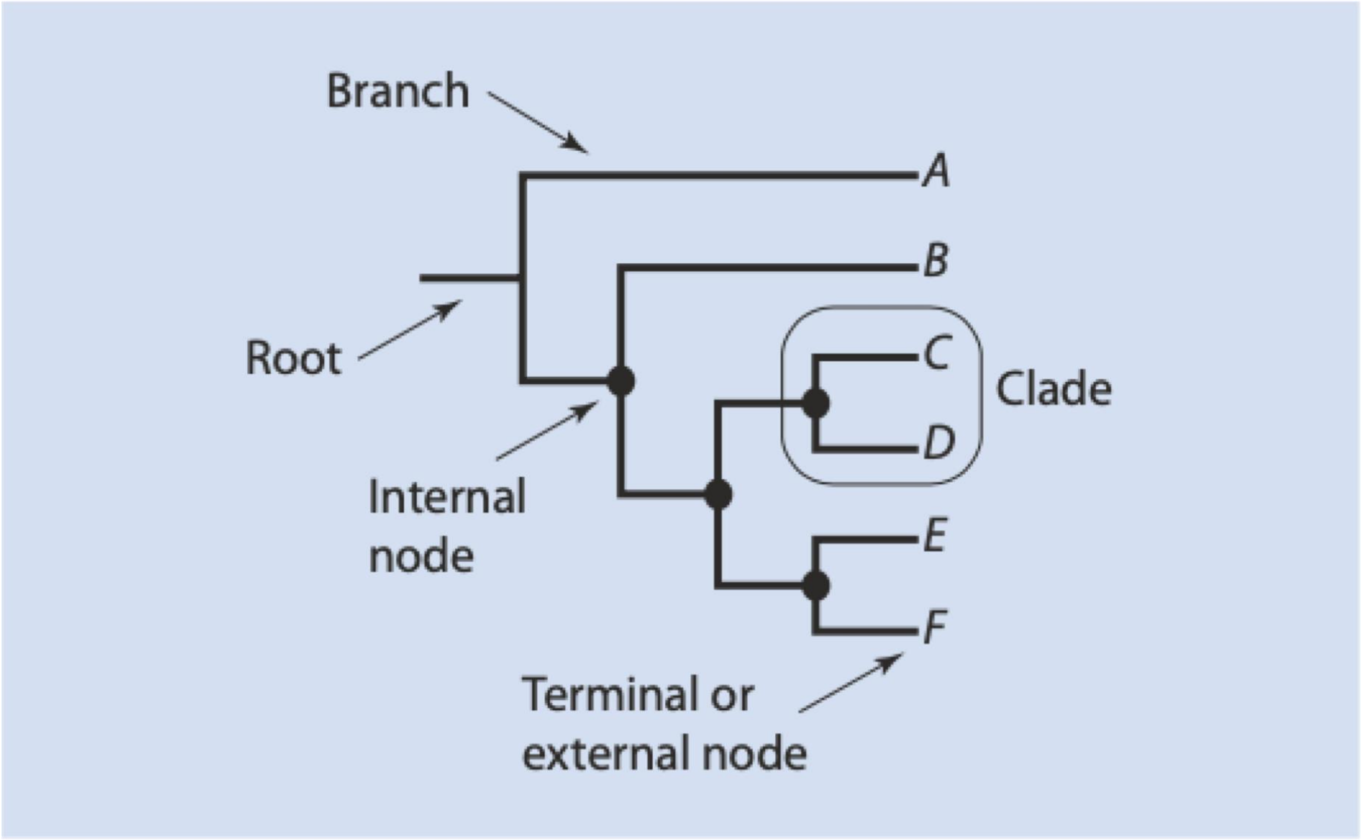
- Objects are referred to as the **taxa** (singular **taxon**) or **operational taxonomic units** (**OTUs**) and in phylogenies based on sequence data they are the individual genes or proteins.

- Each **node** represents an ancestral OTU. The **root** is the common ancestor of all the OTUs. Taxa that occur at the **external nodes** are **leaves**.



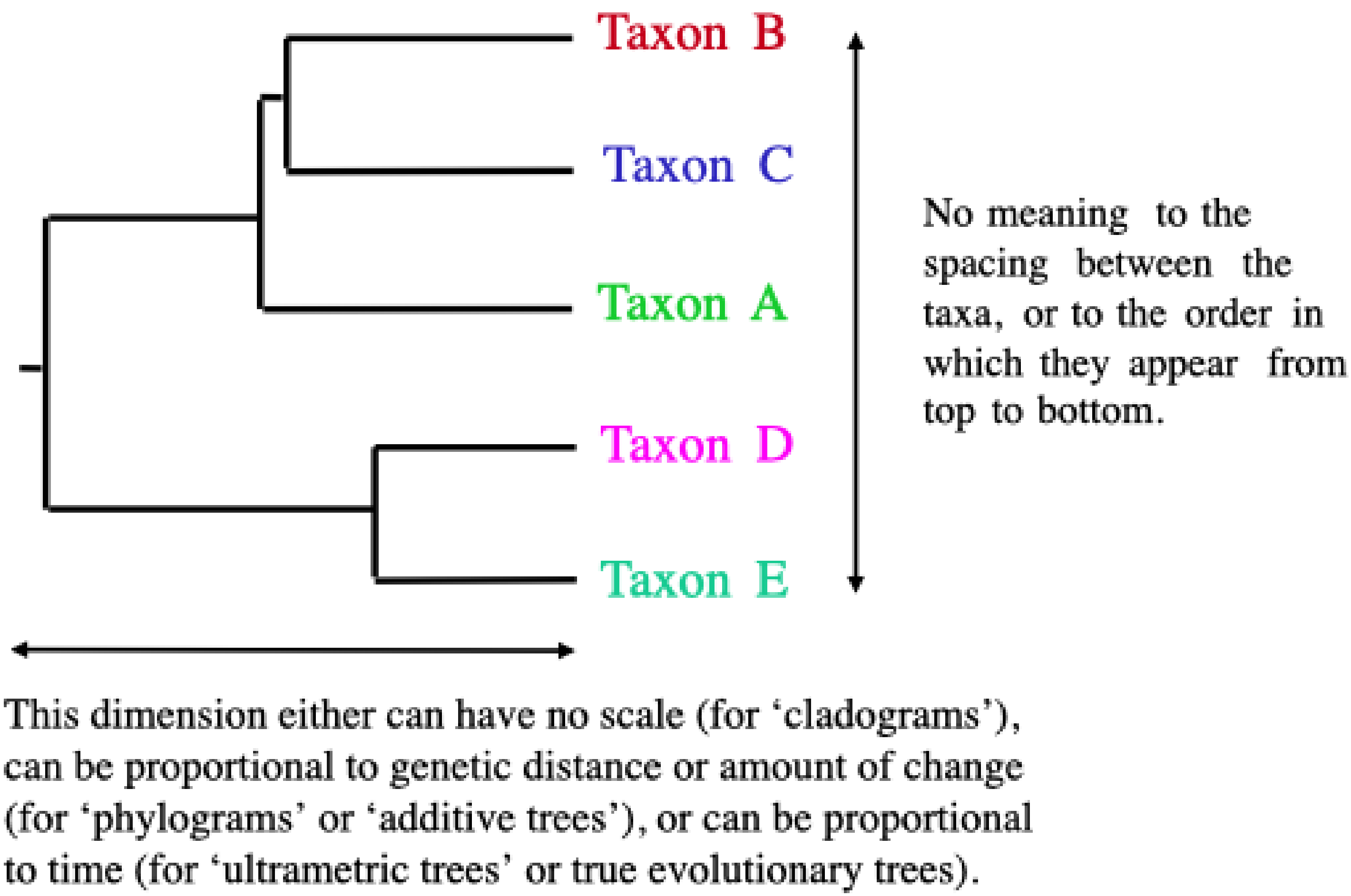
- A **clade** is a group of OTUs that includes several taxons and their common ancestor nodes.

- A **branch** defines the relation between a clade or an OTU and the rest of the tree.



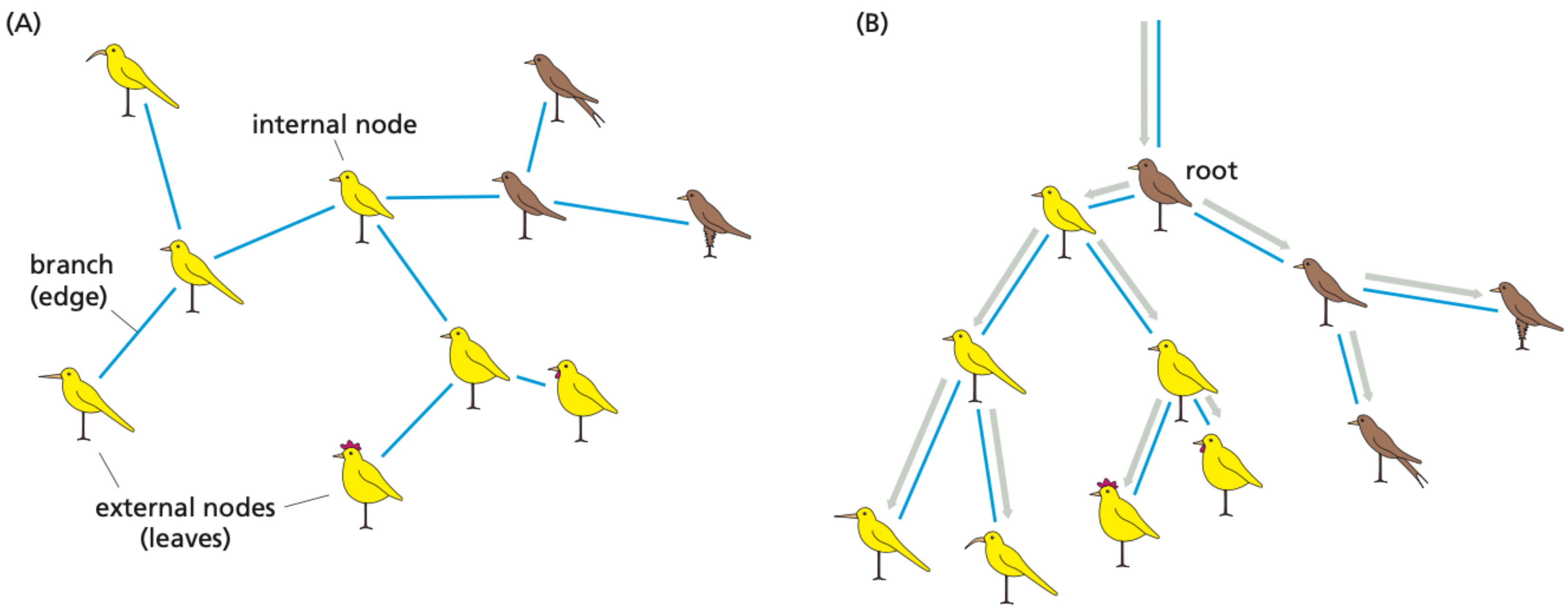
Phylogenetic Trees

- A tree can be **scaled** (in which case the branch length means something in terms of evolutionary time) or **unscaled** (in which case only the topology of the tree is informative).



Phylogenetic Trees

- A tree may be **rooted** or **unrooted**. **Unrooted trees do not identify evolutionary paths.** If your tree is unrooted, you accept the possibility that any of the nodes or OTUs could be closer to the common ancestor than to any of the other OTUs.



- **Rooted** trees represent the divergence of a group of related specied from their last commons ancestor (the root), bu sucessive branching events over time. **Unrooted** trees show the evolutionary relationship between taxa but do not identify their last common ancestor.

Phylogenetic Trees

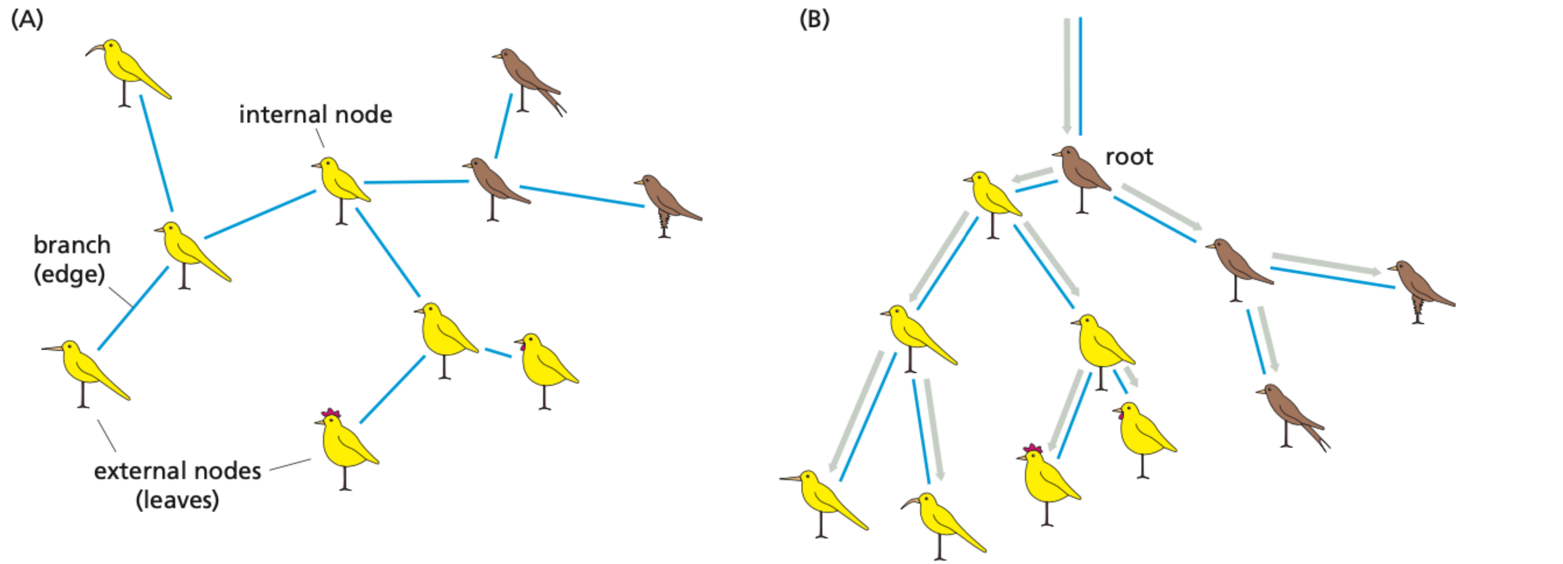
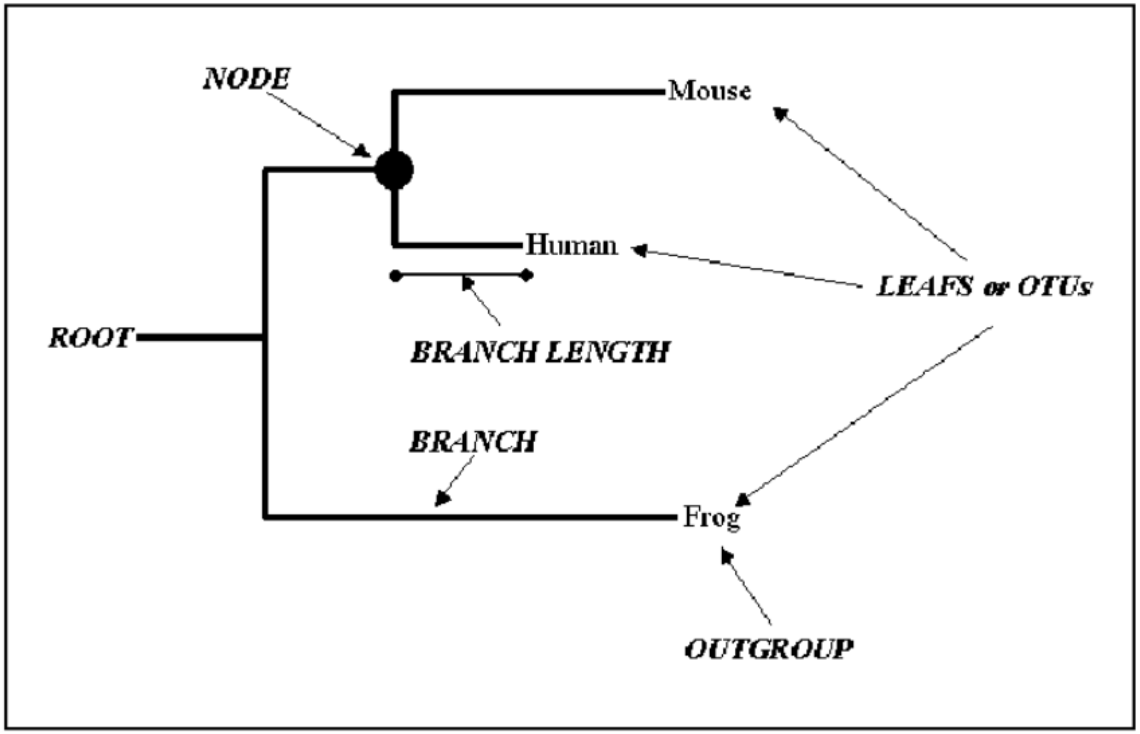


Figure 7.1
Unrooted and rooted phylogenetic trees. These trees reconstruct the evolutionary history of a set of six imaginary extant bird species, which are shown at the outermost tips of the branches—the external nodes or leaves. These are the species from which the data to construct the tree have been taken. These data could be morphological data or sequence data. The birds shown at the internal nodes are the predicted (reconstructed) extinct ancestor species. (A) A fully resolved unrooted tree. This tree is fully resolved in that each internal node has three branches leading from it, one connecting to the ancestor and two to descendants. However, the direction of evolution along the internal branches—that is, which ancestral species has evolved from which—remains undetermined. Thus we cannot distinguish from this tree alone whether the yellow birds evolved from a brown bird, or vice versa. (B) A fully resolved rooted tree for the same set of existing species. The brown bird marked "root" can now be distinguished as the last common ancestor of all the yellow and brown birds. The line upward from the root bird indicates where the ancestors of the root bird would be. In a rooted tree, there is a clear timeline (shown as a gray arrow) from the root to the leaves, and it is clear which species has evolved from which. Thus, the yellow birds did evolve from a brown bird. Apart from the region around the root in (B), the two trees are identical in the relationship between the taxa and give the same information. The arrangement of the branches in space is different, but the information in a phylogenetic tree is contained solely in the branch connections and branch lengths.

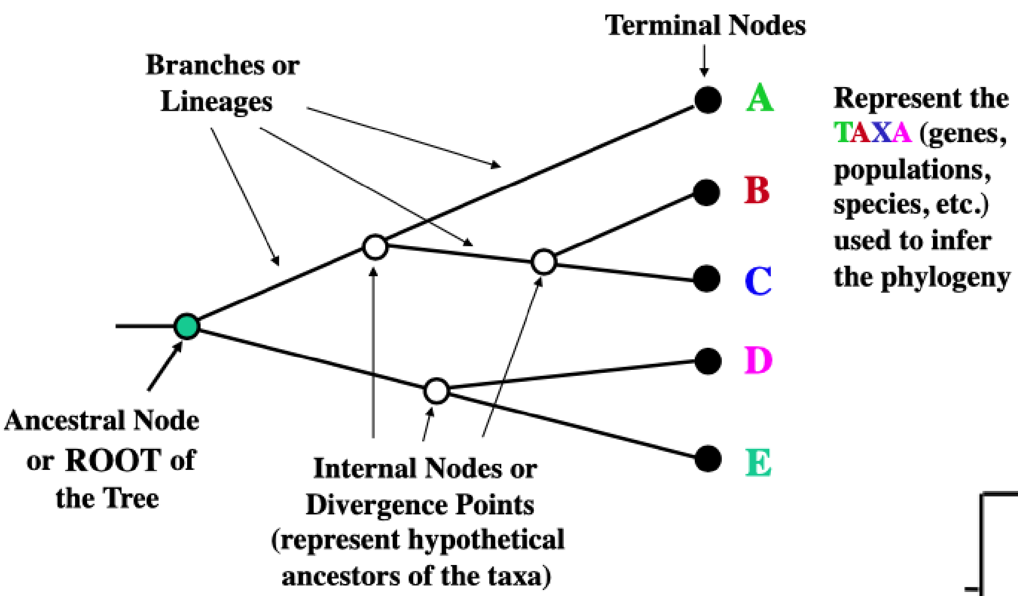
Phylogenetic Trees

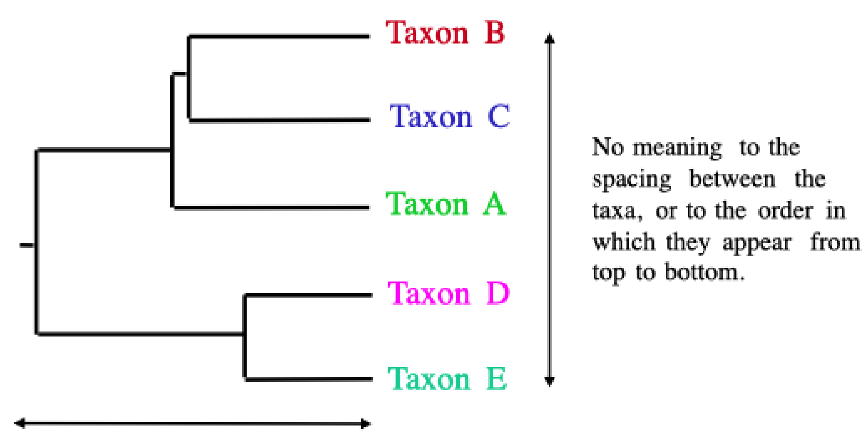
- To help root a tree, we can use an **outgroup** – an organism so distantly related to the others that it can safely be used as a root.

- **For instance, if you are studying primates, you might include a mouse.** Since the **mouse is** known to fall **outsider the primate group**, the branch learning to the mouse helps locate the root: it gives a “reference point”.



Phylogenetic Trees





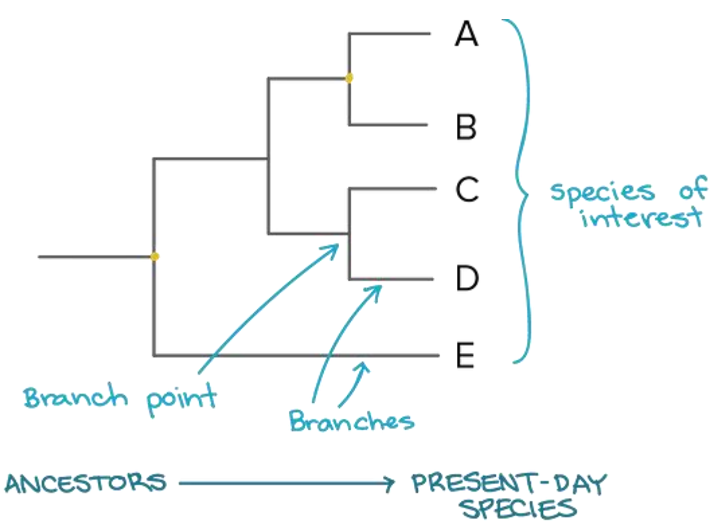
This dimension either can have no scale (for ‘cladograms’), can be proportional to genetic distance or amount of change (for ‘phylograms’ or ‘additive trees’), or can be proportional to time (for ‘ultrametric trees’ or true evolutionary trees).

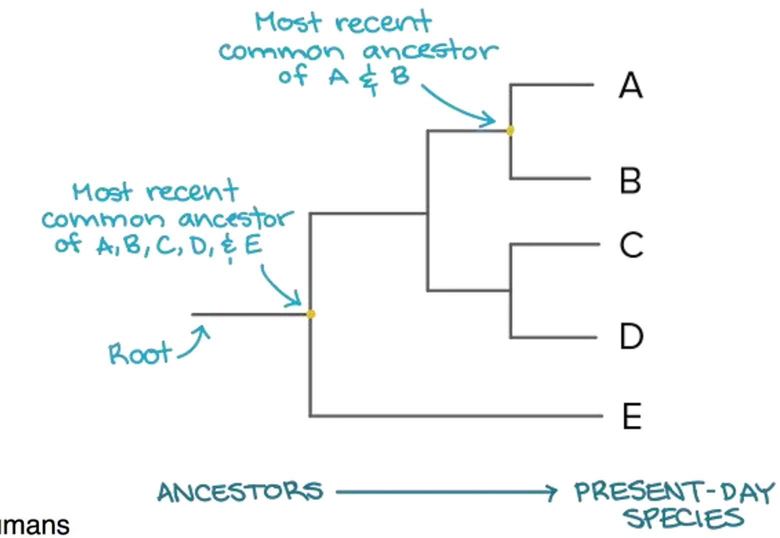
((A,(B,C)),(D,E)) = The above phylogeny as nested parentheses

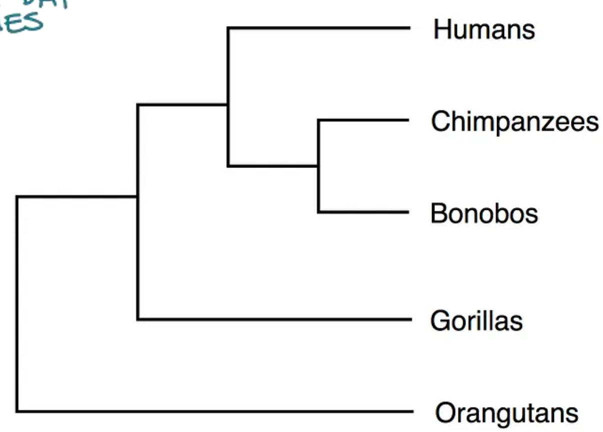
These say that B and C are more closely related to each other than either is to A, and that A, B, and C form a clade that is a sister group to the clade composed of D and E. If the tree has a time scale, then D and E are the most closely related.

Phylogenetic Trees

- Species or groups of interest are found at the tips of the lines called **branches**. Two species are **more related if they have a more recent common ancestor** and less related if they have a less recent common ancestor.

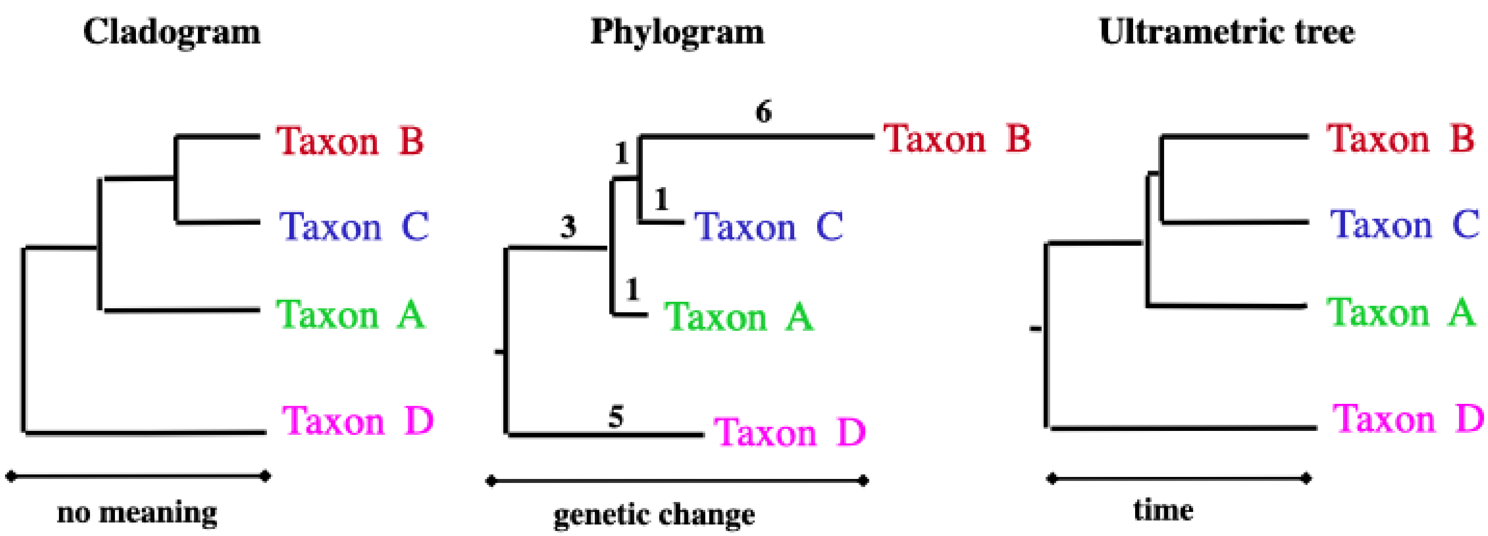






Types of Phylogenetic Trees

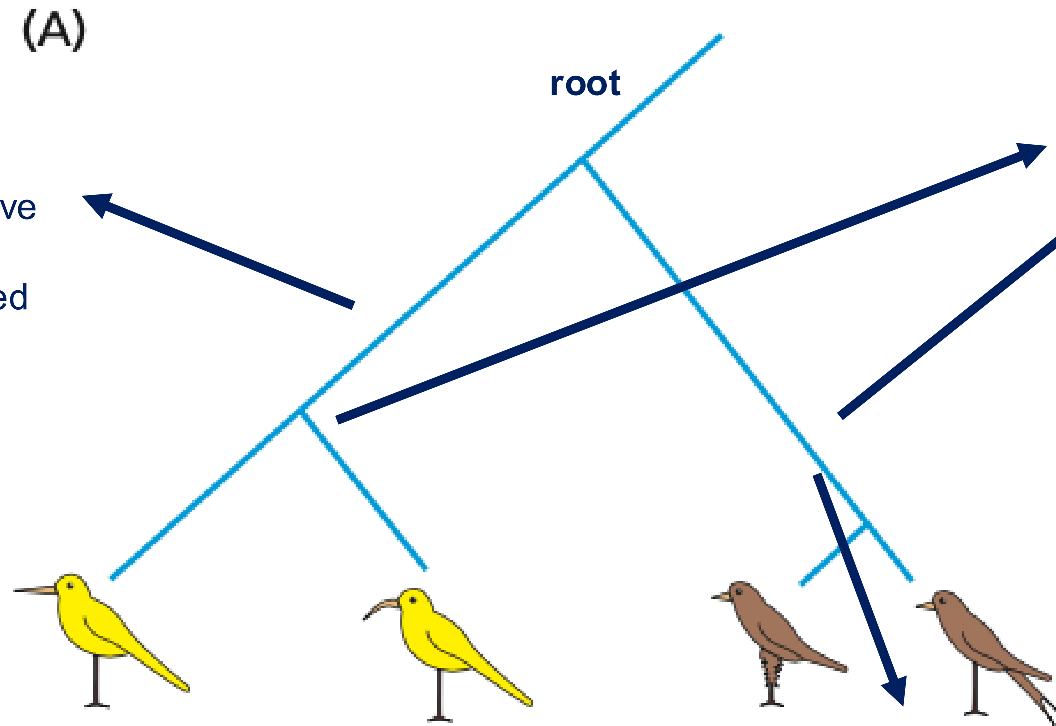
- There are three basic types of tree representation: **cladograms**, **additive trees**, and **ultrametric trees.**



All show the same evolutionary relationships, or branching orders, between the taxa.

Types of Phylogenetic Trees

Cladogram: shows the genealogy of the taxa but **says nothing about the timing** of their divergence.



branch lengths have no meaning, only **topology** is defined

Ancestors are not shown, only the branches (this is a common way to presente the tree, with the ancestors only implied by the internal nodes).

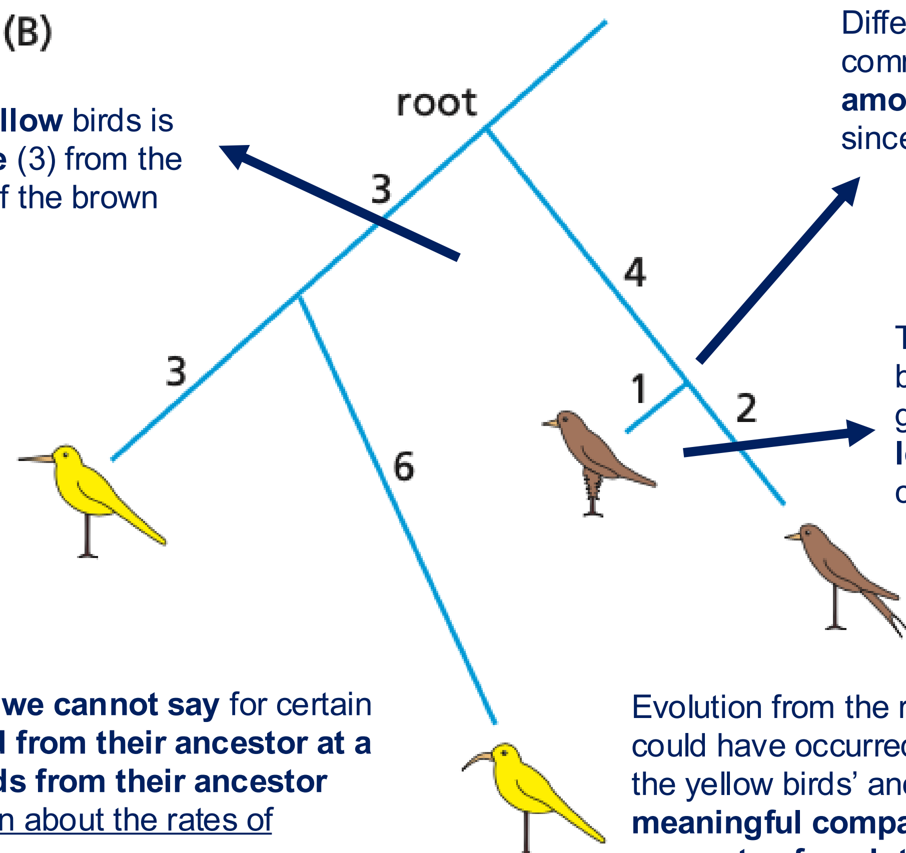
The cladogram tells us that **the four taxa** (the four birds) **have a common ancestor** that initially evolved into two descendants (one common ancestor of the yellow birds and the other common ancestor of the brown birds).

Although the tree has been drawn with different branch lenghts, so that the **brown birds appear to have divergerd more recently than the yellow birds, this is only for artistic effect**: the branch lengths have no scientific basics.

Types of Phylogenetic Trees

Additive Tree: branch lengths represent a **<u>quantitative measure of evolution</u>**, such as being proportional to the number of mutations that have occurred.

The common **ancestor of the yellow** birds is a **smaller evolutionary distance** (3) from the root than the common ancestor of the brown birds (4).



Different branch lengths from a common ancestor indicate **different amounts of evolutionary change** since the ancestor.

The **evolutionary distance** between any two taxa is given by the **sum of the lengths of the branches** connecting them.

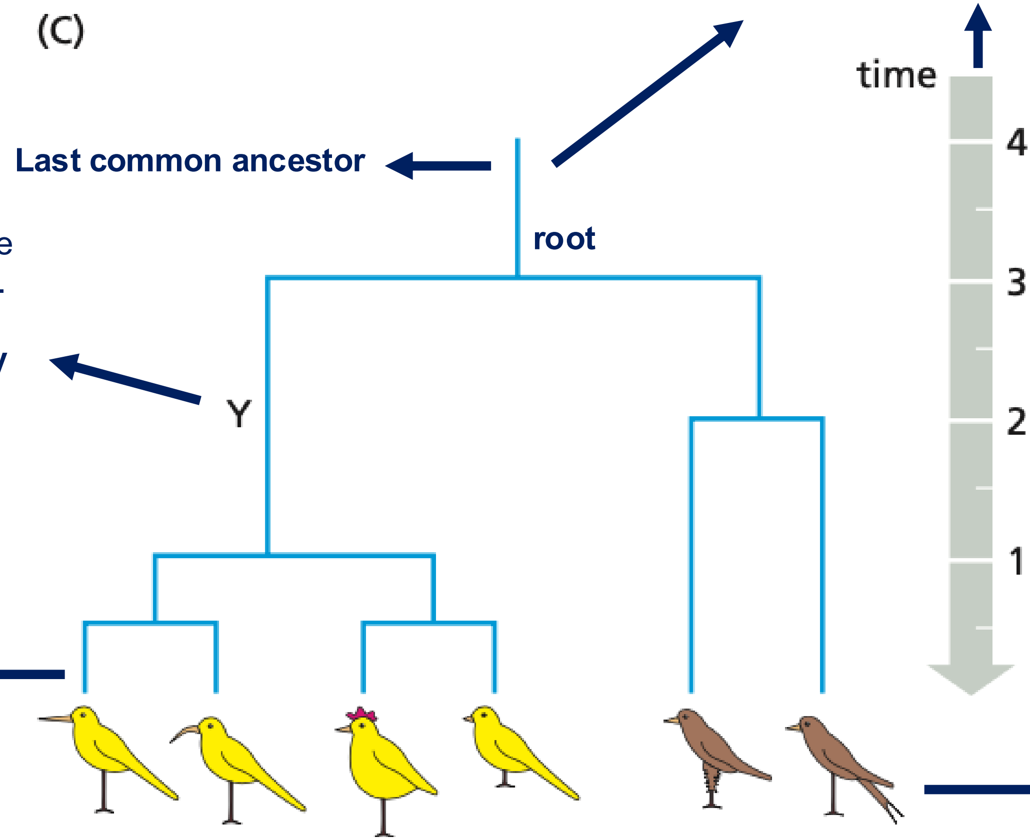
From the representation alone, **we cannot say** for certain **that the brown birds diverged from their ancestor at a later time than the yellow birds from their ancestor** because <u>we have no information about the rates of mutation along the branches!</u>

Evolution from the root to the brown birds' ancestor could have occurred at a much faster rate than to the yellow birds' ancestor. **We can only make meaningful comparisons of the relative amounts of evolutionary divergence.**

Types of Phylogenetic Trees

Ultrametric Tree: in addition to the properties of the additive tree, has the **same constant rate of mutation assumed along all branches**. This is referred to as a **<u>molecular clock.</u>**

All ultrametric trees **have a root**, and one **axis** is directly **proportional to time.**



You could swap the branches in Y, which swaps the birds that will be drawn next to brown birds. As long as **branch lengths and connectivity are mantained**, the trees will have the **same meaning**.

The evolutionary **distance** from a common **ancestor** to all its **descendants** is the **same**.

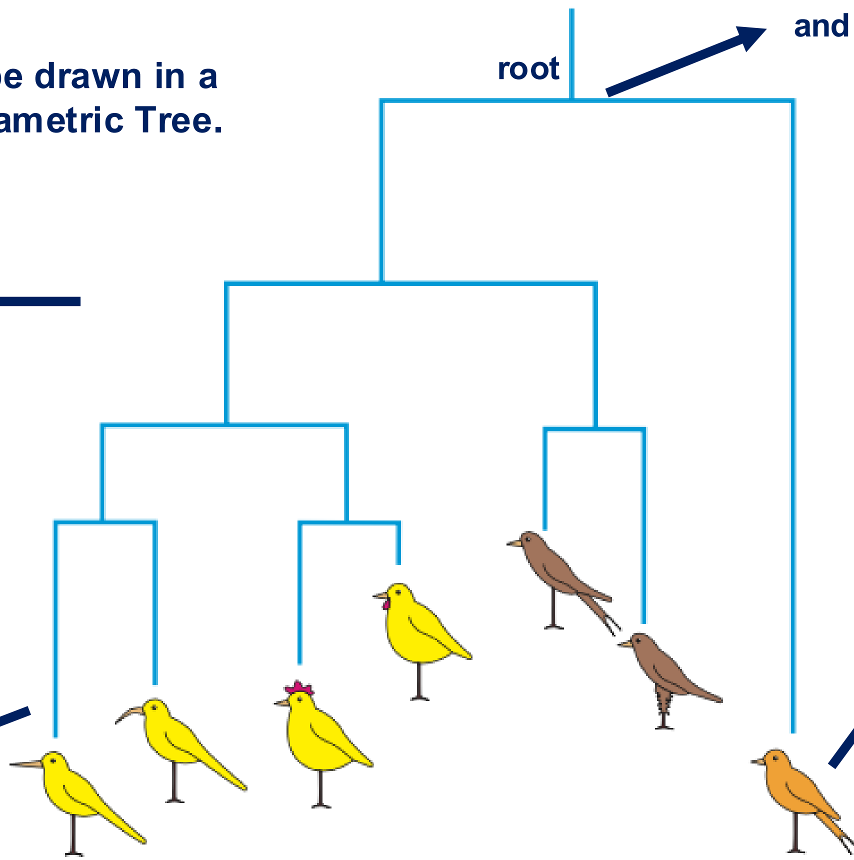
Present Day

Types of Phylogenetic Trees

- Additive trees **often lack a root.** When a rooted tree is required, the most accurate method for placing a root is to **include in the dataset a group of homologous sequences** from species/genes that are **only distantly related** to the main set of species/genes under study **(outgroup)**.

(D)

The Additive Tree can be drawn in a similar style as the Ultrametric Tree.



Root located **between outgroup and the remainder** of the taxa.

The vertical axis is a **measure of evolutionary events**, now proportional to accepted **mutation events instead of time**.

There is **no constant molecular clock**, so leaves will not occur at the same distance from the root.

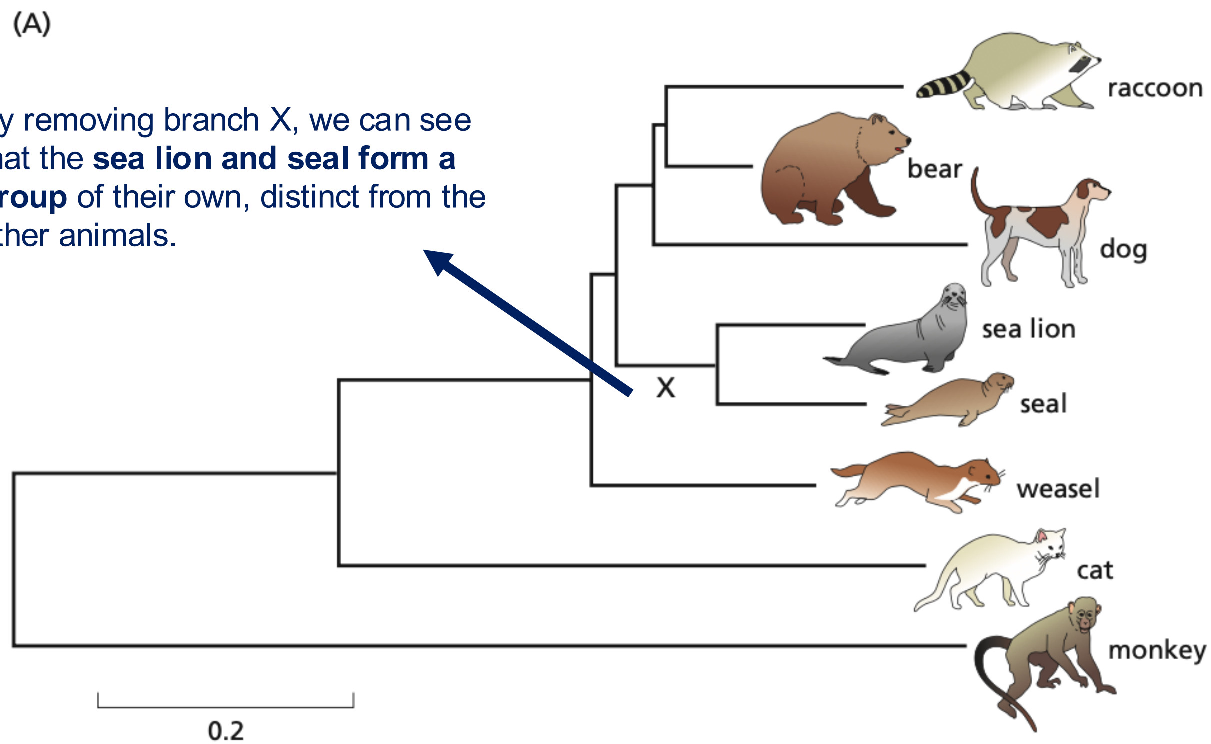
An **outgroup** was used to root the tree, adding the orange bird.

Tree topology

- We need was of describing tree **topology** (the branching pattern a tree, i.e., who is connected to whom) in a form that makes it possible to **compare the topologies** of different trees.

- The graphical views of trees are convenient for human visual interpretation, but not for other tasks such as comparison. One way to sumarize basic information about a tree in **computer-readable format** is to **subdivide** or **split** it into a **collection of subgroups.**

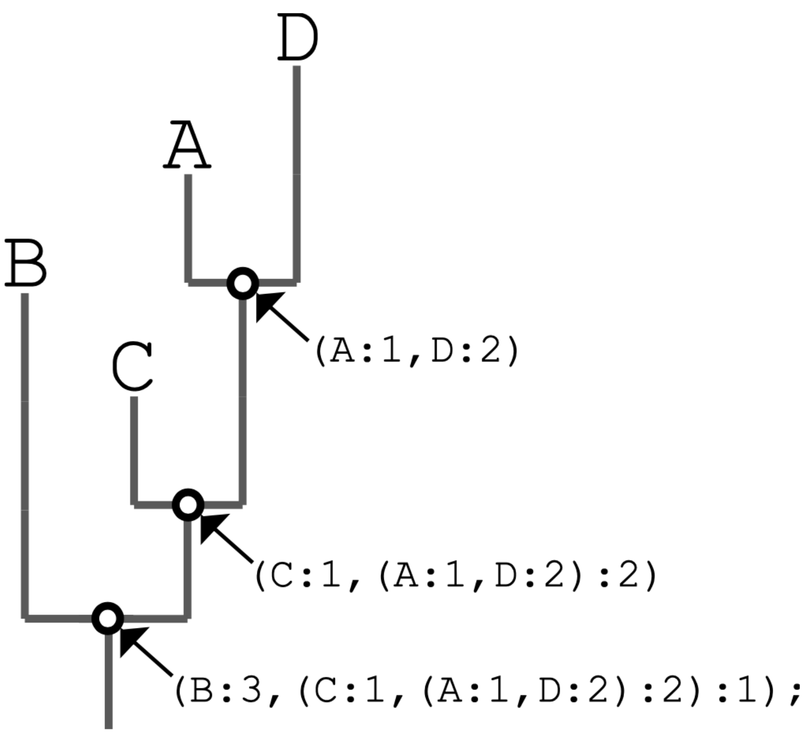
By removing branch X, we can see that the **sea lion and seal form a group** of their own, distinct from the other animals.

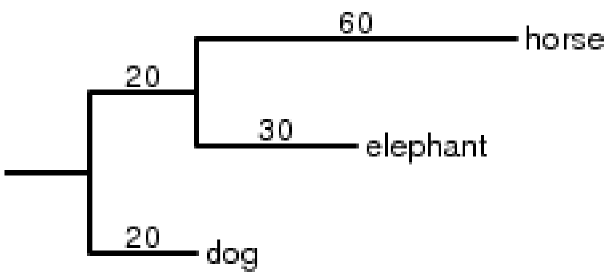


This tree is:
- **Additive**
- **Unrooted**

Newick format

- The complete list of splits of a tree can be written in computer-readable form using the **Newick format.** Tthe **semicolon (;)** at the end completes the Newick format, indicating the end of the tree. Information about the branch lengths can be added, **branch distances** follow **colons (:)**.






```
(dog, (elephant, horse));
```

```
(dog:20, (elephant:30, horse:60):20);
```

```
(B, (C, (A, D)));
```

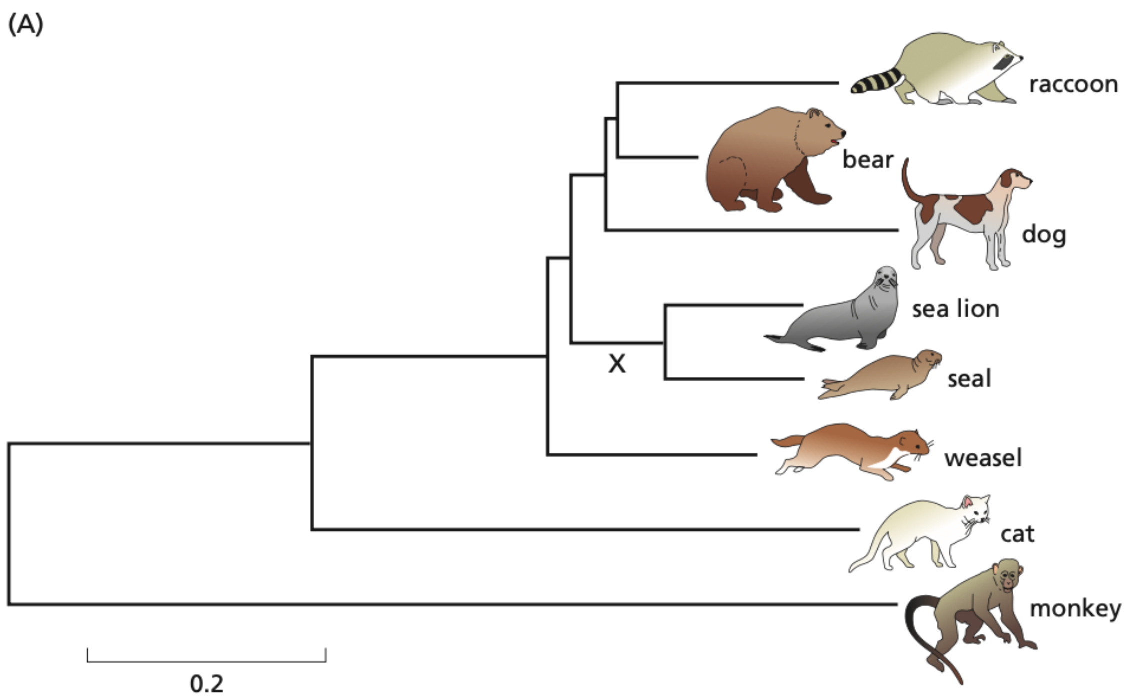
```
(B:3, (C:1, (A:1, D:2):2):1);
```

More information, such **as branch lengths**, can be added to this format.

Newick format

- Each split is written as a **bracketed list** of the taxa as in **`(sea_lion, seal)`**. A complete tree can be described in a similar fashing that identifies every split:

```
((racoon, bear), ((sea_lion, seal), ((monkey, cat), weasel)), dog);
```

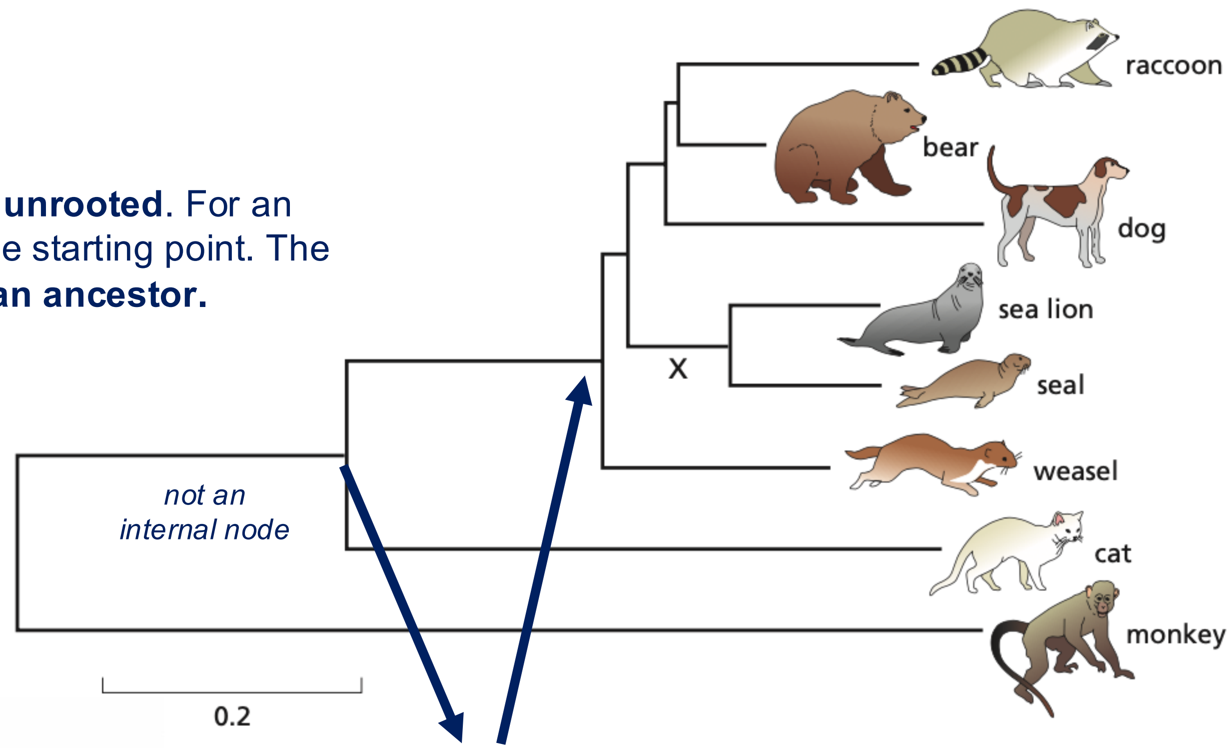


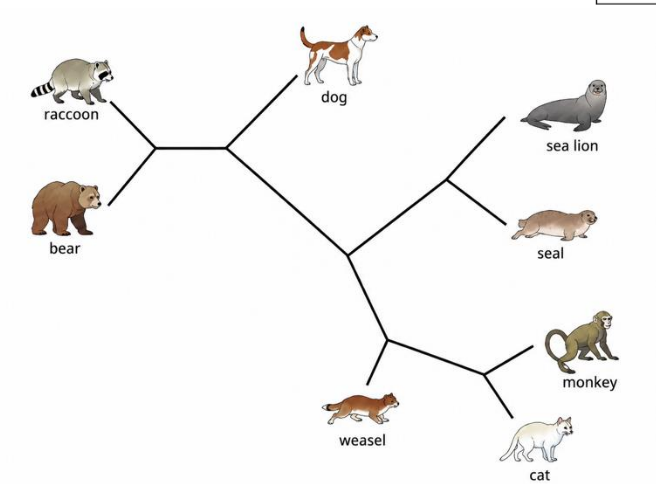

```
((raccoon:0.20, bear:0.07):0.01, ((sea_lion:0.12, seal:0.12):0.08, ((monkey:1.00,
cat:0.47):0.20, weasel:0.18):0.02):0.03, dog:0.25);
```

Newick format

This tree is:
- **Additive**
- **Unrooted**

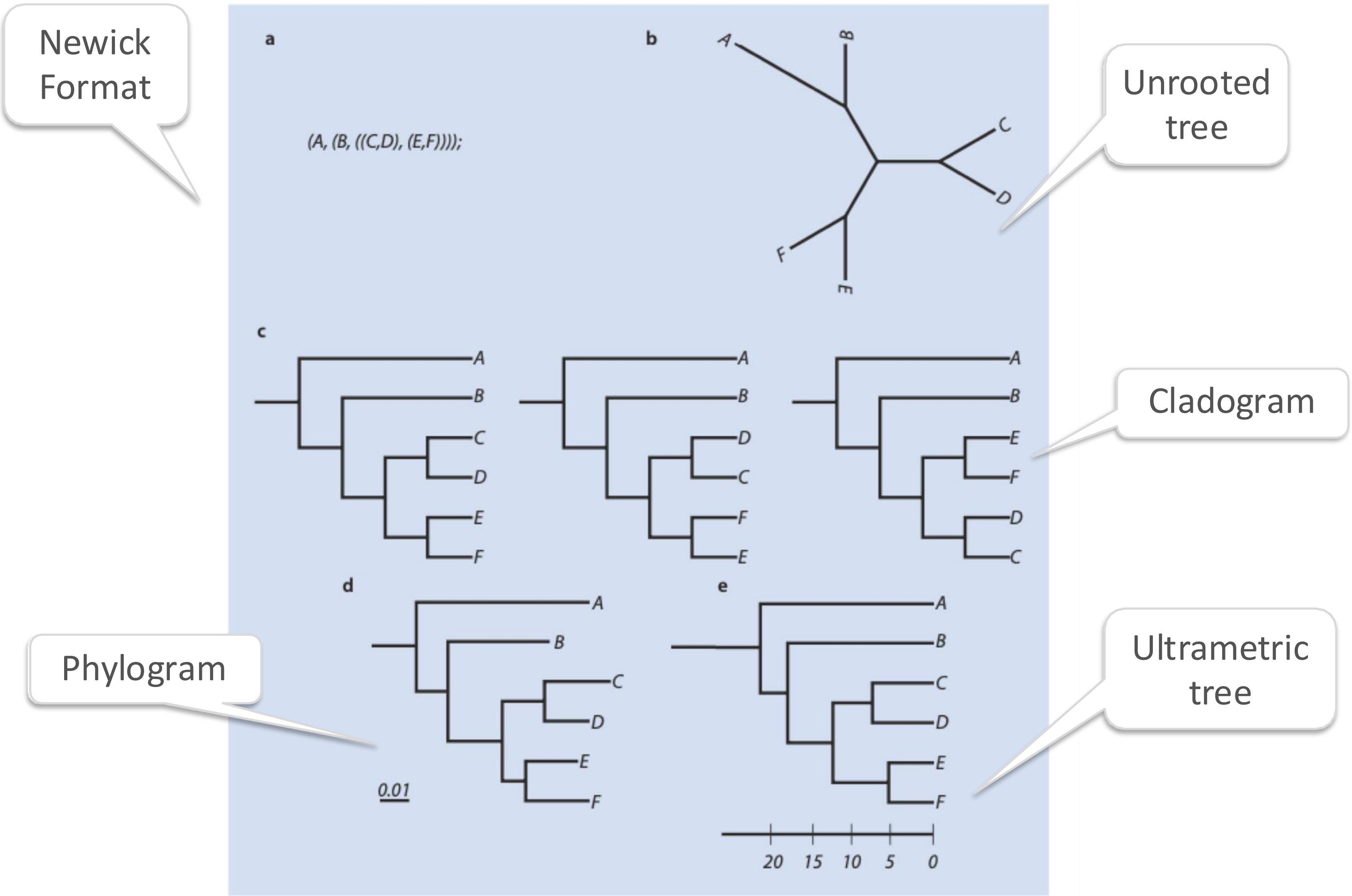
The tree looks rooted, but is **unrooted**. For an unrooted tree, there is no true starting point. The left side is **not necessarily an ancestor.**





Because the tree is unrooted, the branch connecting the monkey to the rest of the tree is **not an internal branch**. Thus, the evolutionary distance from the monkey to its nearest internal node is represented by the sum of the lengths of the two horizontal lines connected by the leftmost vertical line in the figure. **Monkey is attached to its nearest internal node together with cat. In the unrooted topology, monkey and cat are the closest neighbours.**

Tree terminology



Phylogenomics: An Introduction, Christoph Bleidorn, 2017

How to build a phylogenetic tree?

- **Step 1:** Make a multiple alignment from DNA, RNA or aminoacid sequences

```
GGATGCAACTGGTAGTCCCGCGGACGGCTATGCTAGTCTAATCTCTGGCG
AGATGCAACTAGTTGTCTCGCGGACGGC--TGCTAGTCCATCT------A
AGAGGCAGCTGGTTGTCCCACAGACGGCCATGCTAGACCGGTTTCTACAA
AGAGGCACCTGGTTGTCCCGCAGACGGCCATGCTAGACCAGTTTCTACAA
------------TAACATGCGGCACGCGCATGCTAGTCCAATCGAAATCG
------------TAACATGCGGCACGCGCATGCTAGTCCAATTGAAATCG
------TAATATAAGGCACTAGCATGCTTGACGGAGTCCAATGGAGTTCC
------TAATATAAGGCACGCGCCTGCT------AGTCTAATGGAATTCG
```

- **Step 2:** Choose what methods to use and calculate the distance or use the result depending on the method

```
Sequence 1  ATCTATAGCGCGTAT
Sequence 2  AACTATACCGCGCAT
Sequence 3  GTCTGTGGCGCGTAA
Sequence 4  GTTTGTGGCGCGTAA
Sequence 5  GTCTCTGGCGAGTAA
```

	S6	S7	S8	S9	S10
Sequence 6	–	0.9	0.5	0.4	0.3
Sequence 7	0.9	–	0.4	0.3	0.2
Sequence 8	0.5	0.4	–	0.9	0.8
Sequence 9	0.4	0.3	0.9	–	0.7
Sequence 10	0.3	0.2	0.8	0.7	–

Character based vs. **Distance based**

- Use aligned sequences directly
- Sequence data transformed into pairwise distances

Figure 9.5 Character-based versus distance-based phylogenetic methods. Character-based methods such as Maximum Parsimony and Maximum Likelihood use aligned sequences directly during tree inference, while distance-based methods such as Neighbor-Joining first transform the sequence data into pairwise distances.

Building Phylogenetic Trees

- Numerous algorithms have been proposed for reconstructing phylogenetic trees from a multiple sequence alignment. **Methods can be divided into two broad groups:**

  - **Those that first derive a distance measure from each aligned pair of sequences and then use these distances to obtain the tree;**

  - Those that **use the multiple alignment directly,** simultaneously evaluating all sequences at each alignment site, taking each site separately.

- Commonly used distance-based methods are **UPGMA** (Unweighted Pair-Group Method Using Arithmetic Averages) and **NJ** (Neighbor-Joining).

- **Distance methods tend to produce a single tree**, whereas other methods often produce a set of trees that then have to be compared and evaluated using additional techniques in order to choose the one most likely to be correct.

Generating Single Phylogenetic Trees

- Methods based on **distances** (**clustering** methods) gradually build up the tree, starting from a small number of sequences and adding one sequence at each step.

- **Clustering** methods have as strengths their **speed** and **robustness** when compared to methods that search tree topology, but also come with **weaknessess**: there is no associated measure of fitness of the tree to the data, so **trees reconstructed with diferent distance corrections cannot be easily compared.**

- Most clustering methods reconstruct a phylogenetic tree for a set of sequences based on their **pairwise evolutionay distances**. These distances are **straightforward to obtain** but there are potential problems:

  - Alignments can contain errors, some assumptions may not hold for a particular dataset (i.e., identical rates of change at all sites).

How to build a phylogenetic tree?

For **distance-based** methods:

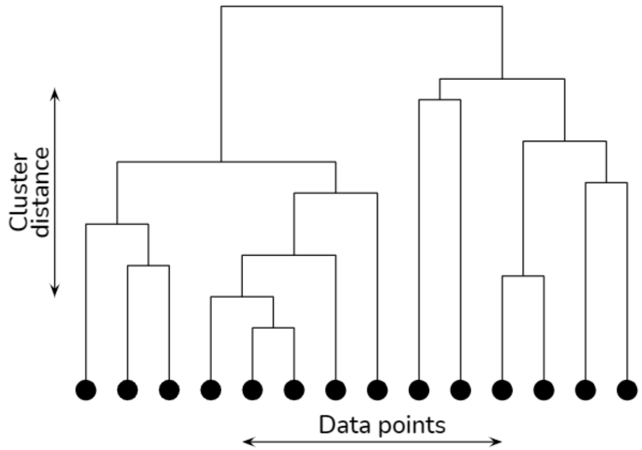
- We need to calculate the distances between all leaves (*taxa*)
- Based on the distance, construct a tree

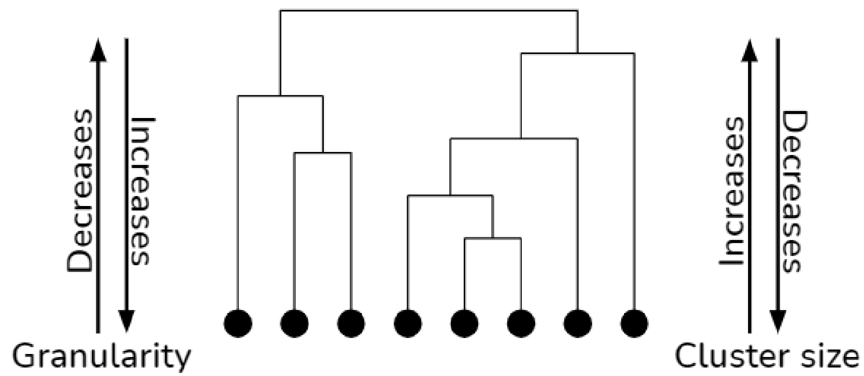
Possible algorithms:

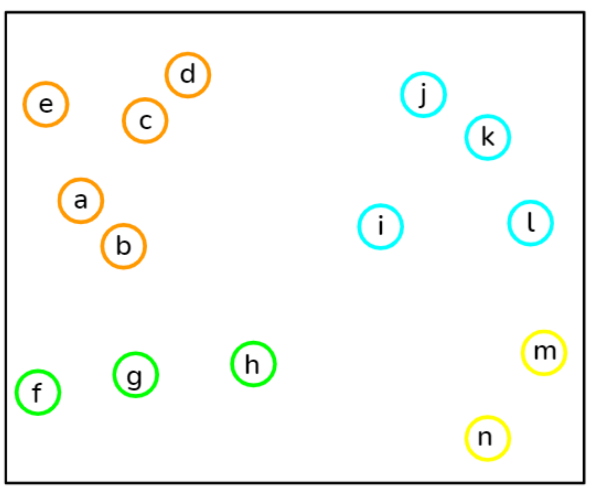
- **UPGMA:** Unweighted Pair Group Method with Arithmetic Mean
- **Neighbour-Joining Method**

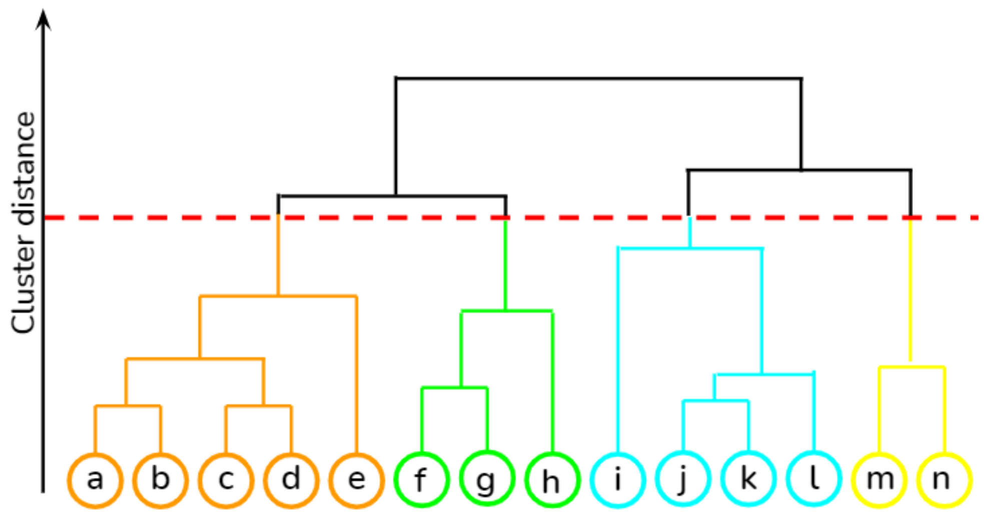
UPGMA

- UPGMA (Unweighted Pair Group Method Using Arithmetic Averages) is based on **aglomerative hierarchical clustering.**









UPGMA

- The UPGMA model assumes a constant **molecular clock** and produces a **rooted and ultrametric tree,** and all leaves are at the same distance from the root.

- The **molecular clock hypothesis dictates that the same number of mutations have occurred** in each sequence **since the time of the last common ancestor**. Thus, the **distance from any node to any leaf that is its descendant will be the same** for all descendants.

- The two sequences with the **shortest evolutionary distanc**e between them are assumed to have been the **last to diverge**, and must therefore have arisen from the **most recent internal node** in the tree.

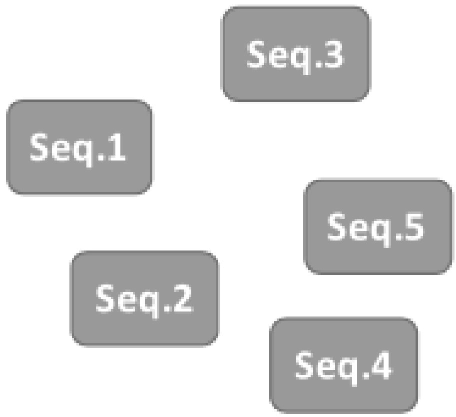
- Furthermore, **their branches** must be of equal length, and so must be **half their distance.**

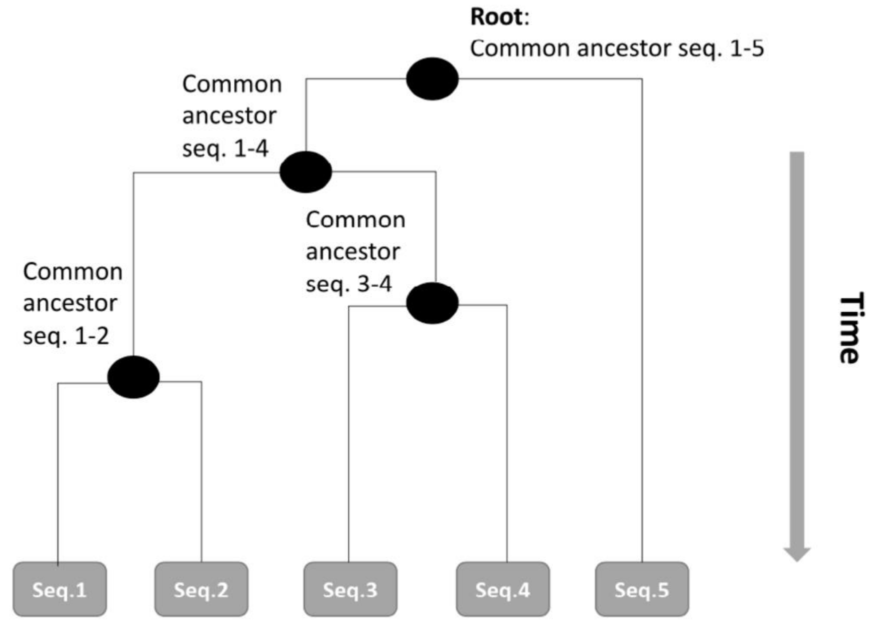
- The sequences are grouped into clusters and the tree is constructed. Initially, all sequences define their own cluster and at each stage, the most **similar clusters are combined.** The tree is complete when all sequences belong to the same cluster.

UPGMA

How to construct a tree with UPGMA?

- Prepare a distance matrix;
- Cluster a pair of leaves (*taxa*) by their **shortest distance (so, the most similar);**
- Compute the average distance between the new cluster and the other taxa, and make a new distance matrix;
- **Repeat until all sequences are within a single cluster, the root node of the tree.**



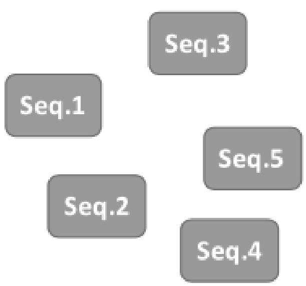


UPGMA

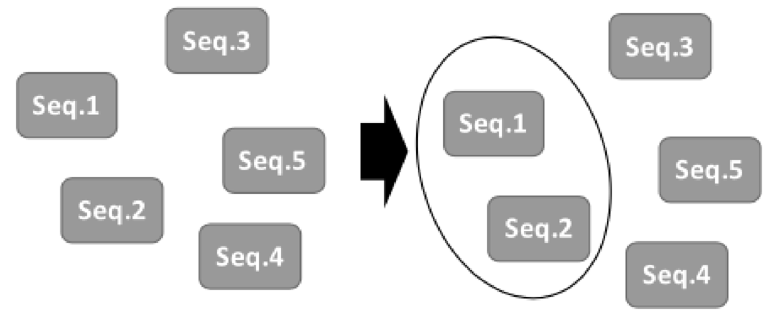
- Each sequence (tree leaf) is first treated as its own cluster (each leaf is a cluster). **They are put at heigh zero in the tree.**

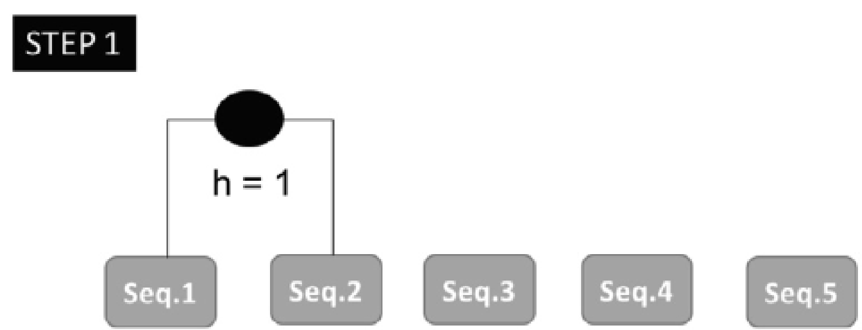
- Then, the algorithm **considers the pair of closest sequences/clusters** (*i.e., checking the minimum value in the distance matrix*) and **joins similar sequences in a internal node.**

- **The internal node has a height in the tree equal to half of the distance between these sequences.**



	S1	S2	S3	S4
S2	$\boxed{2}$			
S3	5	4		
S4	7	6	4	
S5	9	7	6	3





UPGMA

- The new cluster (Seq 1 and 2 in our example) are now a **new cluster.**
- **The distances between all other sequences and the new cluster {Seq1, Seq2} needs to be computed to update the distance matrix.**

	S1	S2	S3	S4
S2	2			
S3	5	4		
S4	7	6	4	
S5	9	7	6	3

	S1, S2	S3	S4
S3	4.5		
S4	6.5	4	
S5	8	6	3

- For instance, the distance between S3 to {S1, S2} is:

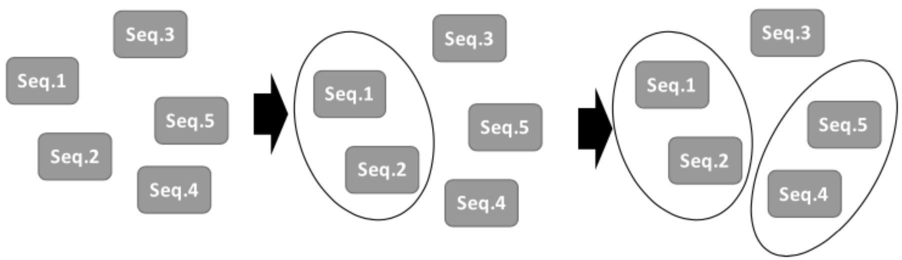
[(S3_to_S1) + (S3_to_S2)] / 2 = [5 + 4] / 2 = 9 / 2 = 4.5

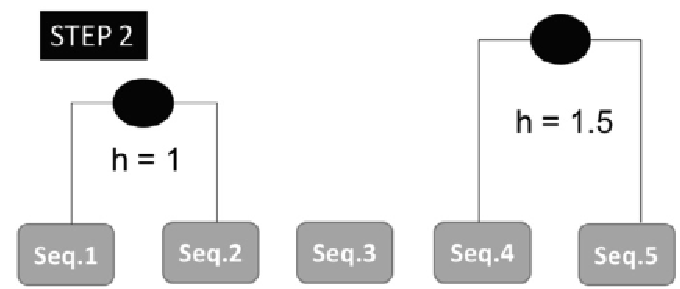
- **From the updated distance matrix, what should we do next?**

UPGMA

	S1	S2	S3	S4
S2	2			
S3	5	4		
S4	7	6	4	
S5	9	7	6	3

	S1, S2	S3	S4
S3	4.5		
S4	6.5	4	
S5	8	6	$\boxed{3}$





	S1	S2	S3	S4
S2	2			
S3	5	4		
S4	7	6	4	
S5	9	7	6	3

	S1, S2	S3	S4
S3	4.5		
S4	6.5	4	
S5	8	6	3

D	S1, S2	S3
S3	$\boxed{4.5}$	
S4, S5	7.25	5

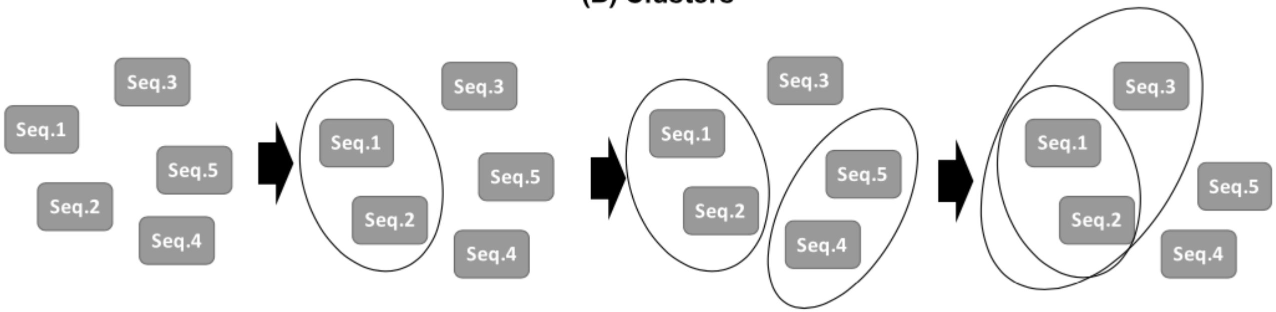
UPGMA

	S1	S2	S3	S4
S2	2			
S3	5	4		
S4	7	6	4	
S5	9	7	6	3

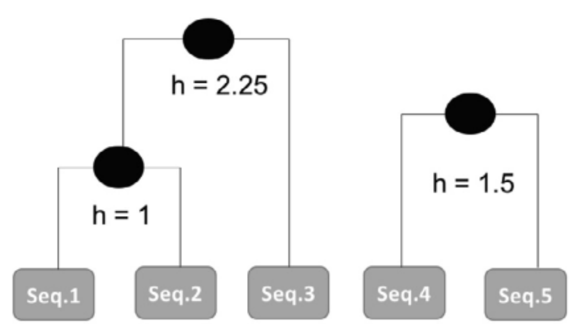
	S1, S2	S3	S4
S3	4.5		
S4	6.5	4	
S5	8	6	3

D	S1, S2	S3
S3	4.5	
S4, S5	7.25	5

(B) Clusters



STEP 3



UPGMA

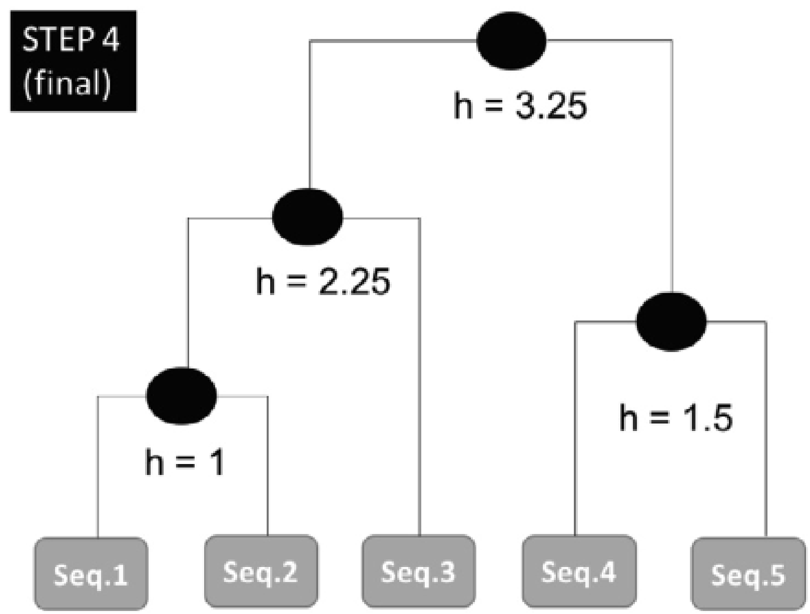
(A) Distance matrix

	S1	S2	S3	S4
S2	2			
S3	5	4		
S4	7	6	4	
S5	9	7	6	3

	S1, S2	S3	S4
S3	4.5		
S4	6.5	4	
S5	8	6	3

D	S1, S2	S3
S3	4.5	
S4, S5	7.25	5

D	S1, S2,S3
S4, S5	6.5



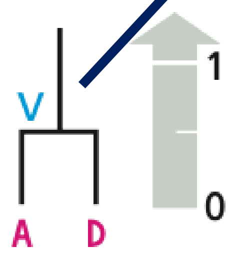
UPGMA

A and D are the clostest

They form cluster **V = {A, D}**

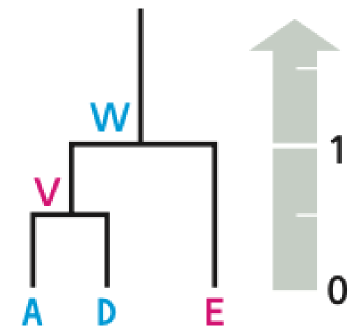
(A)

d_{ij}	A	B	C	D	E	F
A	–	6	8	1	2	6
B		–	8	6	6	4
C			–	8	8	8
D				–	2	6
E					–	6



(B)

d_{ij}	B	C	E	F	V
B	–	8	6	4	6
C		–	8	8	8
E			–	6	2
F				–	6



Their distance is 1 and the branches need to be equal, so ½ = 0.5 (same branch length)

Now cluster **E and V** are the closest. Their distance is 2, so respective branches are **at length 2/2 = 1.**

Now cluster E and V are the closest. But how to compute the distance between two clusters?

UPGMA

The distance between two clusters is defined as follows. Consider the construction of a tree for N sequences and suppose that at some stage you have clusters X containing N_X sequences and Y containing N_Y sequences. Initially, each cluster will contain just one sequence. The evolutionary distance (d_{XY}) between the two clusters X and Y is defined as the arithmetic average of the distances between their constituent sequences, that is

$$d_{XY} = \frac{1}{N_X N_Y} \sum_{i \in X, j \in Y} d_{ij} \qquad \text{(EQ8.15)}$$

where i labels all sequences in cluster X, j labels all sequences in cluster Y, and d_{ij} is the distance between sequences i and j. When two clusters X and Y are combined to make a new cluster Z there is an efficient way of calculating the distances of other clusters such as W to the new cluster. The new distances can all be defined using existing cluster-to-cluster distances without the need to use the constituent sequence-to-sequence distances, using the equation

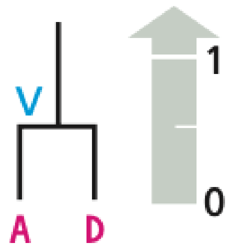
$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y} \qquad \text{(EQ8.16)}$$

UPGMA

Now cluster E and V are the closest. But how to compute the distance between two clusters?

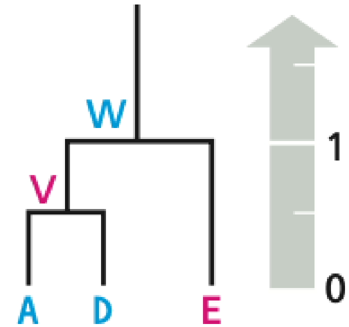
(A)

d_{ij}	A	B	C	D	E	F
A	–	6	8	1	2	6
B		–	8	6	6	4
C			–	8	8	8
D				–	2	6
E					–	6



(B)

d_{ij}	B	C	E	F	V
B	–	8	6	4	6
C		–	8	8	8
E			–	6	2
F				–	6



$$d_{XY} = \frac{1}{N_X N_Y} \sum_{i \in X, j \in Y} d_{ij}$$

Arithmetic average of the distances between their constituent sequences:

$$d_{EV} = 1 / (N_V * N_E) * (d_{AE} + d_{DE}) = ½ * 4 = 2$$

$$d_{EV} = (d_{AE} + d_{DE}) / 2$$

$$d_{EV} = (2 + 2) / 2 = 2$$

$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

$$d_{EV} = [(N_A * d_{AE}) + (N_D * d_{DE})] / [N_A + N_D]$$

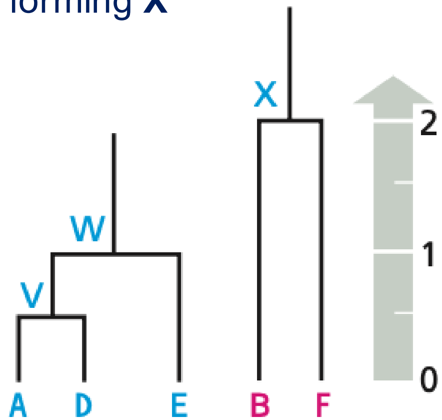
$$d_{EV} = [(1 * 2) + (1 * 2)] / [1 + 1] = 4/2 = 2$$

UPGMA

(C)

B and F are now the closest, forming **X**

d_{ij}	B	C	F	W
B	–	8	4	6
C		–	8	8
F			–	6



$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

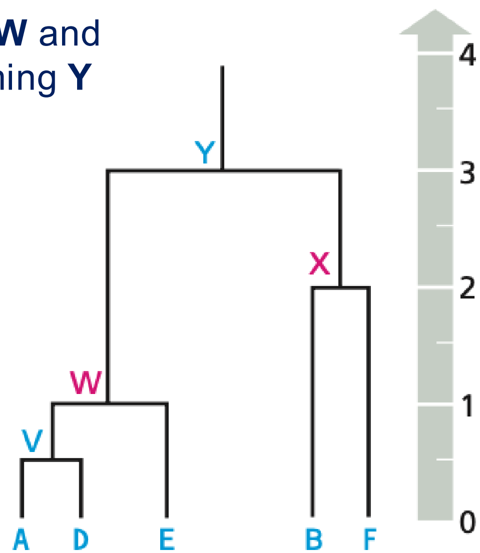
$$d_{XW} = [(N_B * d_{BW}) + (N_F * d_{FW})] / [N_B + N_F]$$

$$d_{XW} = [(1 * 6) + (1 * 6)] / [1 + 1] = 6$$

(D)

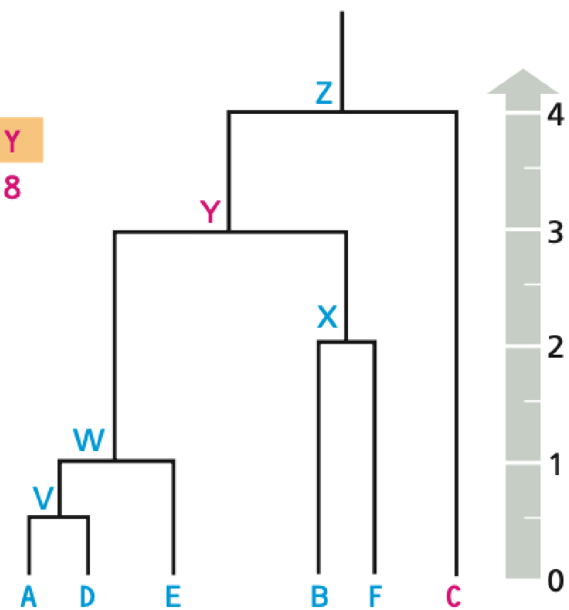
Then, **W** and **X**, forming **Y**

d_{ij}	C	W	X
C	–	8	8
W		–	6



(E)

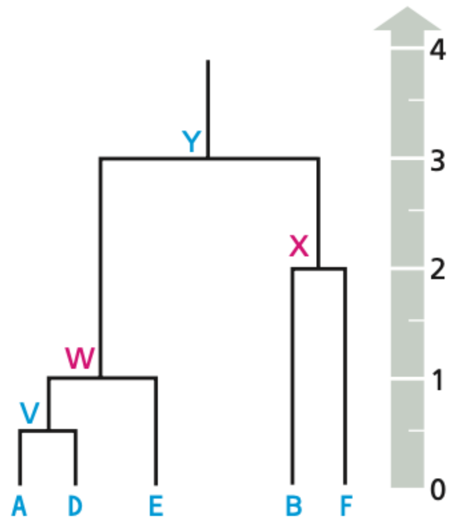
d_{ij}	C	Y
C	–	8



UPGMA

(D)

d_{ij}	C	W	X
C	–	8	8
W		–	6

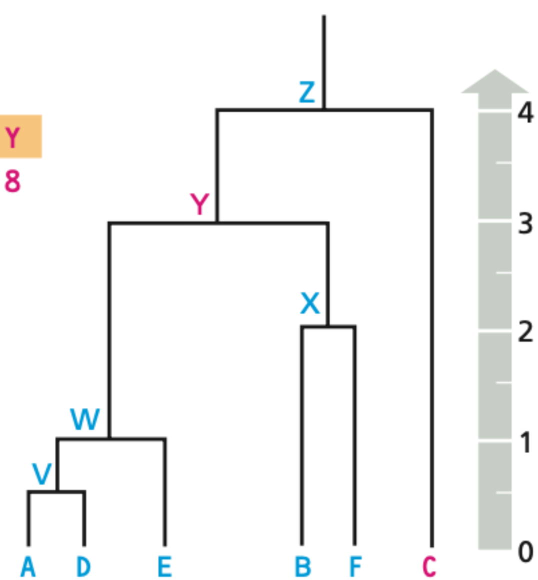


$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

$d_{YC} = [(N_W * d_{WC}) + (N_X * d_{XC})] / [N_W + N_X]$

(E)

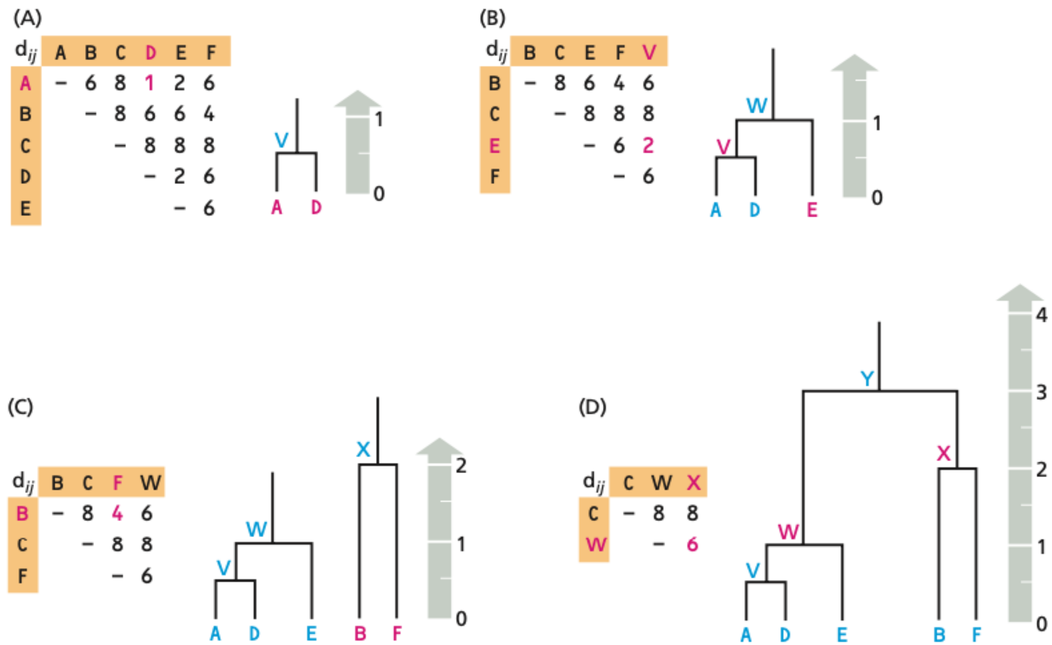
d_{ij}	C	Y
C	–	8



$d_{YC} = [(3 * 8) + (2 * 8)] / [3 + 2]$

$= [\ 24 + 16] / 5 = 40 / 5 = 8$

UPGMA



(E)

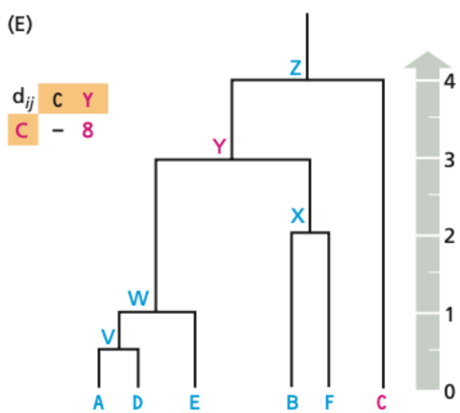


Figure 8.3
A worked example of the UPGMA method of phylogenetic tree reconstruction for six sequences A to F. (A) The initial evolutionary distance (d_{ij}) matrix (left) showing that sequences A and D are closest. They are selected in the first step to produce internal node V (right). The evolutionary distance is shown by the gray arrow. (B) The distance matrix including node V (left), from which it can be deduced that V and E are closest, resulting in internal node W (right). (C,D) Subsequent steps defining nodes X, Y, and Z and resulting in the final tree (E).

UPGMA vs NJ

- **UPGMA** makes the assumption that the **sequences evolved at a constant equal rate** (the **molecular clock** hypothesis), and produces **rooted ultrametric trees** with all sequences at the same distance from the last common ancestor.

- The **NJ** method belongs to the group of **minimum evolution methods**, and produces **unrooted additive trees**. It is derived from the assumption that the most suitable tree to represent the data will be that which proposes the **least amount of evolution**, measured as the total branch length in the tree.

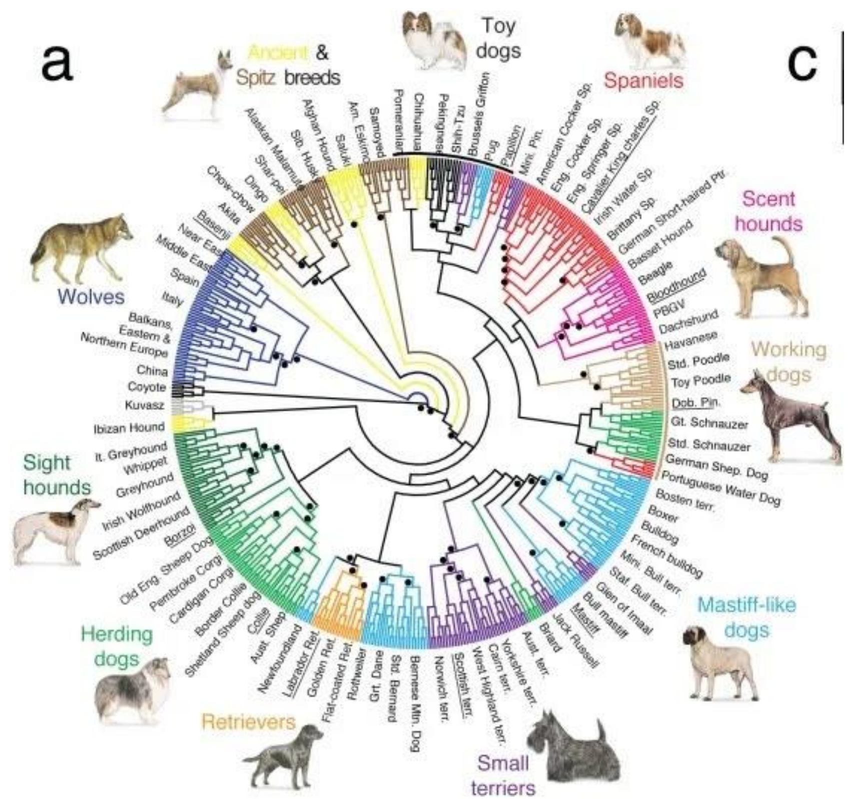
UPGMA vs. NJ

UPGMA is quite simple and popular, but has many limitations:

- Assumes that the mutation rate is uniform (**molecular clock**) and therefore the trees are *ultrametric*). If this is the case, then UPGMA returns the optimal solution. However, **this assumption is rarely true in practice.**

An alternative is **Neighbour Joining (NJ)** that does not assume the constant mutation rates across the different lineages.

- NJ creates **<u>unrooted trees</u>** and does not require data to be *ultrametric*.
- The criterion to select the clusters to merge takes into account the **distances between clusters** but also **seeks to select clusters pairs where the nodes are apart from other nodes**.
- **Clusters to be merged are selected based on a trade-off**: seeking pairs of nodes that have a short distance among themselves and large differences from other clusters.



- *BioPython* features for phylogenetic analysis are implemented mainly in the ***Bio.Phylo*** **module.**

- It implements data structures to represente phylogenetic trees and methods to load, save, and explore these trees.

- **Consider that you are given 5 DNA sequences with their labels:**

```
Alpha     AACGTGGCCACAT
Beta      AAGGTCGCCACAC
Gamma     CAGTTCGCCACAA
Delta     GAGATTTCCGCCT
Epsilon   GAGATCTCCGCCC
```

- You are given the task of constructing the phylogenetic tree to represent these sequences **based on distance-based phylogenetic inference methods.**

BioPython Functions for Phylogenetic Analysis

- We will be using the **`DistanceTreeConstructor`** for this task. The **`DistanceTreeConstructor`** supports two algorithms: `UPGMA` and `NJ`. You can download the `msa.phy` file which contains the input sequences at: *https://github.com/biopython/biopython/blob/master/Tests/TreeConstruction/msa.phy*

```
# Import modules
from Bio import Phylo
from Bio.Phylo.TreeConstruction import DistanceCalculator
from Bio.Phylo.TreeConstruction import DistanceTreeConstructor
from Bio import AlignIO

# Read the sequences and align
aln = AlignIO.read('msa.phy', 'phylip')
```

```
5 13
Alpha      AACGTGGCCACAT
Beta       AAGGTCGCCACAC
Gamma      CAGTTCGCCACAA
Delta      GAGATTTCCGCCT
Epsilon    GAGATCTCCGCCC
```

The first line is a header:

- The first is the **number of organisms;**
- The second is the **length of each DNA sequence;**
- The **.phy** extension indicates that the file contains phylogenetic data.

- Phylogenetics algorithms require the sequences to be aligned to build the tree. The first step involves **importing the already aligned sequences** into the program, using **`AlignIO`** to read the data and store it in a variable `aln`.

```
# Import modules
from Bio import Phylo
from Bio.Phylo.TreeConstruction import DistanceCalculator
from Bio.Phylo.TreeConstruction import DistanceTreeConstructor
from Bio import AlignIO

# Read the sequences and align
aln = AlignIO.read('msa.phy', 'phylip')

# Print the alignment
print(aln)
```

```
Alignment with 5 rows and 13 columns
AACGTGGCCACAT Alpha
AAGGTCGCCACAC Beta
CAGTTCGCCACAA Gamma
GAGATTTCCGCCT Delta
GAGATCTCCGCCC Epsilon
```

- Now we use a distance matrix to store information on how similar each DNA sequence is from every other sequence. The more similar, the more likely they are to be evolutionarily close. We create a `DistanceCalculator` with the `'identity'` model and store the distance matrix of the alignment in a variable `dm`.

```
# Calculate the distance matrix
calculator = DistanceCalculator('identity')
dm = calculator.get_distance(aln)

# Print the distance Matrix
print('\nDistance Matrix\n===================')
print(dm)
```

`'identity'`: distance is equal to the proportion of non-matching nucleotides

```
Distance Matrix
===================
Alpha   0.000000
Beta    0.230769    0.000000
Gamma   0.384615    0.230769    0.000000
Delta   0.538462    0.538462    0.538462    0.000000
Epsilon 0.615385    0.384615    0.461538    0.153846    0.000000
            Alpha       Beta        Gamma       Delta       Epsilon
```

- Finally, you can use the `DistanceTreeConstructor` to create a `constructor`. Then, calling the `upgma` method allows you create the phylogenetic tree according to UPGMA algorithm and store it in a variable `tree`, which you can print with `Phylo.draw_ascii()`.

```
# Construct the phylogenetic tree using UPGMA algorithm
constructor = DistanceTreeConstructor()
tree = constructor.upgma(dm)

# Print the phylogenetic tree in the terminal
print('\nPhylogenetic Tree\n==================')
Phylo.draw_ascii(tree)
```

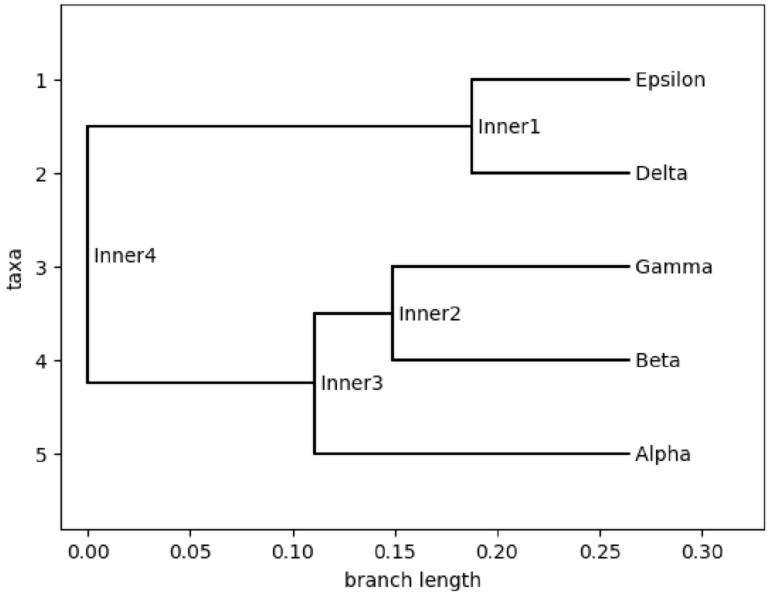
```
Phylogenetic Tree
==================
                          ____________________ Epsilon
  _______________________|
 |                       |____________________ Delta
_|
 |                    ______________________________ Gamma
 |           ________|
 |__________|        |______________________________ Beta
            |
            |________________________________________ Alpha
```

- We can also use `Phylo.draw()` but this requires **`matplotlib` to be installed.**

```
pip install matplotlib
```

```
# Draw the phylogenetic tree
Phylo.draw(tree)
```

```
Distance Matrix
===================
Alpha   0.000000
Beta    0.230769    0.000000
Gamma   0.384615    0.230769    0.000000
Delta   0.538462    0.538462    0.538462    0.000000
Epsilon 0.615385    0.384615    0.461538    0.153846
0.000000
            Alpha       Beta        Gamma       Delta
Epsilon
```



https://biopython.org/docs/latest/Tutorial/chapter_phylo.html

Conclusions

- **Phylogenetic** trees reflect the **evolutionary history** of a group of species or a group of homologous genes. They are constructed using the information contained in **alignments of multiple homologous sequences**.

- To build a tree reflecting the **evolutionary** history of a set of species, the chosen sequences must be **orthologs**. **Gene family** trees can include **any homologous** sequences.

- Trees may also be **rooted or unrooted**, and can be represented as **cladograms**, **additive trees**, or **ultrametric trees (always rooted).**

- There are many numerous methods for tree construction: some work directly from the **MSA**, whereas others use **evolutionary distances** calculated between each pair of sequences in the alignment.

Future Challenges

- Phylogenetic analysis is a powerful tool for answering many types of biological questions. **Yet, they are dynamic construc**ts – depending on the methods used, sequences included, sampling of species, parameters, rooting, and data quality.

- As DNA sequencing technology costs decrease, the challenge in the era of “big data” for phylogenetics and bioinformatics lies on **the availability of individuals with sufficient expertise to perform analyses** (and infrastructure for computations). Bioinformaticians with the skills to carry out phylogenetic analyses of genes, genomes, and proteins will continue to be on demand.

- It’s also crucial to **systematically analyse** the accuracy, sensitivity and specificity of **current tools**, so that the community may decide on the most appropriate for the desired downstream task.

- Going forward, different types of bioinformatic and phylogenetic analysis will provide **new ways to understand the world and teach us new ways to adapt** to our environment.